Weaponizing Interdependence

Master Research Blueprint: Reciprocal Semiconductor Controls, Rare-Earth Supply Resilience, Strategic Opacity, and Endogenous Leverage Erosion

Consolidated research design | synthesis of Drafts 0–5 and the final executive-summary revision

Research information freeze: 11 October 2026 | master synthesis: 2026

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Certification, reading guide, and status

Document type: methodological master blueprint, not an empirical paper and not a claim that numerical results have been obtained. This edition reconciles the uploaded seven iterations and is deliberately self-critical. Each specification is identified as **fixed design**, **conditional theorem**, **to be estimated**, **scenario-only**, or **unverified implementation**. An attractive formula is not an observed result; a source URL is not an extracted dataset; a repository diagram is not working software. No status box in this document implies otherwise.

Research question. *How does the exercise of asymmetric U.S.-led semiconductor controls and Chinese rare-earth restrictions change the opponent's short-run technological or material access, induce or suppress long-run adaptation, and thereby change the equilibrium value and use of economic coercion? When, if ever, does rare-earth diversification cross a durable physical-resilience threshold or a genuinely self-reinforcing economic tipping point?*

Thesis. Coercion can create contemporaneous leverage because a target relies on hard-to-substitute production capabilities. But the same coercion can change future investment, disclosure incentives, policy support, supply-chain topology, and technological progress. The sign and size of that feedback are empirical questions. With positive induced adaptation, a sufficiently patient coercer may optimally restrain current pressure because pressure consumes the future value of the chokepoint. With negative induced adaptation, no general 'self-eroding' conclusion follows; this is a meaningful falsification possibility, not a failure of the research.

Nonnegotiable scope. The empirical chapters are genuinely separate before integration. Rare earths: China versus physically feasible non-Chinese supply for magnet-related elements, tracked by mine/feedstock, oxide separation, metallurgy/alloys, and finished magnets. Semiconductors: U.S.-led and allied controls by item, use case and jurisdiction, differentiating Chinese domestic production capability, foreign supply access, and the combined functional bottleneck. The third component is *an observable policy-opacity covariate*, not a third industrial chokepoint. An independent AI-model/adoption ecosystem game is excluded. Demand from AI-related applications may enter as an exogenous semiconductor demand variable.

Deliverable hierarchy. D1 = clean conceptual theory and complete variable definitions; D2 = source- and vintage-correct data architecture; D3 = project and layer estimates; D4 = properly evaluated probability forecasts; D5 = empirically disciplined dynamic game with verified numerical diagnostics; D6 = reproducible paper. This document aims to settle D1 and specify D2–D6. It does **not** assert that D2–D6 have been executed. A genuine publication-ready empirical paper requires an independently checked dataset, code, results, robustness and references. The completion audit at the end distinguishes those tasks explicitly.

How the drafts were synthesized

Source	Material retained	Material superseded or corrected
Draft 0 (30 pp.)	Structural tipping-versus-threshold taxonomy; logistic positive feedback with critical interaction coefficient 4; network cascades; historical-vintage validation; empirical feasibility discipline.	Financial contagion/bank-crisis case studies are a methodological analogy, not an extra empirical paper.

Source	Material retained	Material superseded or corrected
Draft 1 (45 pp.)	Equal analytical status for both chokepoints; contemporaneous policy chronology; payoff interpretation; realistic limits on structural identification; optional beliefs.	Early game variants with inconsistent state derivatives, ambiguous timing, or unconstrained equilibrium claims.
Draft 2 (43 pp.)	Correct bounded saturation dynamics; one-player coercion benchmark; open-loop versus feedback clarity; independent semiconductor bottlenecks.	Independent clipping of coupled interior Nash solutions; incompletely specified pipelines.
Draft 3 (55 pp.)	Four-state Markov game; project-stage schema; opacity pillars; source vintages and software architecture.	Incomplete physical feedstock routing; invalid backtest endpoints; positivity restrictions left implicit.
Draft 4 (52 pp.)	Explicit nine-regime constrained Nash cases; stepwise physical inputs; full indicator scale; data dictionaries.	Stage-by-stage subtraction treated as one N-1 outage; signed investment pipelines without positivity guard.
Draft 5 (54 pp.)	Single-supplier all-stage removal; legal-vintage table; strong programming and forecasting protocol; academic positioning.	Linear game dynamics inconsistent with their own invariant-domain proof; ambiguous map between physical and game states; unfinished end-to-end executable package.
Executive Summary (11 pp.)	Reminder to present a concise public-facing overview alongside technical report.	Its revised player/state meanings, steady states, fabricated numeric 'equilibria,' loss of definitions, and unverifiable sources are rejected.

Locked design decisions and registered sensitivity choices

Issue	Primary	Alternatives / falsification
Mineral elements	Nd, Pr, Dy, Tb; distinct element identities and grade recipes	Sm/other controlled REEs in a **separate** policy appendix; expanded 17-REE case only if data support it
Critical mineral outcome	Delivered finished-magnet services supported through all upstream stages; report separated-oxide and alloy throughput alongside	Separate oxide-only and Dy/Tb-only adequacy; import-inclusive and full-China-free cases
Resilience jurisdiction	Defined U.S.-aligned importing bloc B; China-origin-controlled inputs removed	Strict non-China global, U.S. only, geographic versus ownership origin
Outage severity	First remove China (strategic coercion scenario); then a separate worst-single-non-China-supplier N-1 test	Country, corporate group, logistics hub, two concurrent failures
Historical panel	Project-stage-quarter with two time clocks (event time; publication date)	Monthly for observed dates; annual if information sparse
Forecast targets	2028Q4, 2031Q4, 2035Q4; 4 consecutive quarters of required service	0.5, 0.75, 0.9, 1.0 service fractions; single-quarter and 8-quarter thresholds
Semiconductor layer domain	Advanced logic production and enabling inputs plus HBM/packaging conditional on target architecture	Mature-node use cases; separate consumer versus high-end computing target
Semiconductor slow game state	Chinese *domestic* functional substitutability of prescribed technology stack	Differentiable surrogate; alternative architecture-specific aggregations

Issue	Primary	Alternatives / falsification
Game state y	Fraction of bloc final magnet-service demand deliverable without Chinese material/processing	Stronger China-free-plus-one-other-supplier outage
Game adaptation delay	First-order bounded Markov pipelines	Two-stage Erlang, fixed-delay history model **only outside** four-state HJB
Positive adaptation	A testable restriction, **not** a default fact	Negative and zero responses explicitly allowed
Opacity	Ten scored observability measures, fixed 40/35/25 pillar weights	Equal weights, validity-calibrated latent factor, policy-sector variants
Game solution	Four-state stationary Markov-perfect equilibrium *candidate* through coupled discounted HJB, active-set control updates, multi-start, refinement	Finite-state mixed strategy; open-loop / constant-policy benchmarks separately
Forecast inference	Cause-specific multistate project hazards with stochastic ramps/network reoptimization	Pooled parametric hazards, Bayesian hierarchical, model averaging
Research freeze	Facts publicly knowable by October 11, 2026	New frozen release after explicit documented source update

1. Theoretical foundation and falsifiable contributions

1.1 Why the two chokepoints are asymmetric

China's advantage in magnet rare earths is much more pronounced in refining and permanent-magnet production than in raw ore extraction. The IEA's 2026 assessment reports approximate Chinese shares of **60% of 2024 magnet-REE mining, 91% of refined output, and 94% of sintered permanent magnets**, and substantially thinner announced non-Chinese pipelines downstream than upstream [R25]. The policy object is not simply a Chinese endowment of minerals: it is a *multi-stage input-processing system with specific chemical, engineering, feedstock and qualification constraints*. Announced capacity in one stage cannot freely replace missing capacity in another. The empirical target is physically feasible service to specified end users under a China-removal stress test, not mining tonnes or a generic supply concentration ratio.

By contrast, U.S.-led semiconductor leverage is a *technology/access network*. It involves semiconductor design and process software, lithography, deposition/etch/implant equipment, inspection and metrology, foundry process mastery, HBM, advanced packaging, American extraterritorial rules and U.S.-person requirements, and controls adopted independently by allies. The BIS December 2024 package, for example, added controls on 24 categories of semiconductor-manufacturing equipment, three types of software, and HBM; Dutch 2025 controls also expanded to some measurement and inspection equipment; Japan's advanced equipment rules cover 23 classes [R26–R29]. A legal licensing requirement should never be automatically coded as a complete embargo. In January 2026 BIS shifted certain advanced-chip licenses to conditional case-by-case review [R30]. Every control variable must be time-varying, product-specific, and as-of-date correct.

This asymmetry shapes the data and mathematics. Rare-earth resilience may be reconstructed from project histories, conversion yields, physical capacity and feasible network throughput. Semiconductor resilience requires *target-specific performance evidence*, distinct from money spent, patents filed, wafers started or overall chip shipment values. Reciprocal-game equations are intentionally lower-dimensional

approximations to these separately measured mechanisms, not exact identities inherited automatically from physical laws.

1.2 The closest existing research and genuine novelty

Farrell and Newman [R01] theorize how central nodes of international networks can become coercive assets. Zou, Poncin and Bertinelli [R04] already formulate a U.S.–China rare-earth differential game. Clayton, Maggiori and Schreger [R02] explicitly model the coercion/integration/security trade-off. Woods [R03] offers a dynamic opacity/signaling framework for rare-earth policy with **illustrative**, not externally estimated, simulation parameters. The semiconductor firm-level literature [R05–R07] does not unanimously establish that restrictions increase actual capabilities: R&D, patenting, learning and operating substitutability may move differently. Consequently this project must **not** claim originality for inventing economic coercion, mineral differential games, or uncertainty itself.

The defensible contribution is a joint empirical-theoretical chain: **(i)** vintage-correct probability distributions of actual non-Chinese project completion and stage-feasible magnet production; **(ii)** distinct estimates of immediate Chinese semiconductor access shocks and lagged indigenous adaptation, by layer; **(iii)** an auditable observability-only measure of policy opacity and tests of investment delay or overreaction; **(iv)** a reciprocal, constrained dynamic game whose state and adaptation parameters are *disciplined* by, but not presumed fully identified from, the empirical chapters; and **(v)** forecasts and policy counterfactuals that quantify the erosion of future leverage under explicit uncertainty.

1.3 Hypotheses, direction of evidence, and null outcomes

H1 - Rare-earth implementation. Verified finance, permitting, engineering readiness, offtake and government support predict commissioning beyond announced nominal capacity and target date. This is primarily a predictive result, not necessarily a causal effect.

H2 - Strategic pressure and diversification. Greater Chinese restriction exposure changes initiation, financing, commissioning or subsequent capacity for non-Chinese projects; the sign and lag may differ by stage and geography.

H3 - Semiconductor first impact. U.S./allied restrictions cause or predict immediate access losses in the targeted layer, conditional on remaining foreign substitutes, stockpiles, exemptions, and shipment adjustment.

H4 - Semiconductor induced capability. Restriction exposure changes subsequent domestic R&D, capex, patents and *independently verified* functional capability. Effects may be positive, zero, or negative, with distinct lag distributions.

H5 - Direct opacity drag. Greater *observable* ambiguity or regulatory nondisclosure, separately from prices, technological uncertainty and geopolitical risk, changes industrial project progression hazards. The positive quadratic/overreaction variant is secondary and may be rejected.

H6 - Endogenous resilience feedback. Existing operating non-Chinese projects help or fail to help subsequent projects pass financing, technical qualification and commissioning barriers. An observed positive interaction is not by itself a bistable phase transition; true multiplicity requires the separate nonlinear test in Section 5.

H7 - Self-eroding leverage. Conditional on positive restriction-induced adaptation and an intertemporal payoff trade-off, forward-looking optimal restriction intensity may be below the equivalent myopic optimum. This is provable in the constant-control benchmark; its generality and magnitude in the full reciprocal game require numerical verification.

H8 - Reciprocal strategic interaction. Estimated or scenario-based contemporaneous complementarity/retaliation can increase restrictions, while greater adaptation responsiveness and strategic patience can offset it. No theorem guarantees universal restraint in every two-player specification.

1.4 What 'tipping' is allowed to mean

We distinguish four phenomena that earlier drafts sometimes conflated. **Physical adequacy threshold:** modeled feasible service crosses, for example, 75% or 100% of required demand; this is a **first-passage/viability** event, not a bifurcation. **Steady-state feasibility threshold:** investment is sufficient to sustain a selected normalized resilience level in a saturating ODE; still not necessarily a bifurcation. **Dynamical tipping:** a feedback parameter creates multiple steady states and an unstable threshold, fold, or hysteresis; this needs a formally derived condition and empirical support. **Network propagation/cascade:** loss of a supplier changes feasible downstream production through physical edges or induces correlated finance decisions; cascading input shortfalls do not establish a macroeconomic critical transition unless additional nonlinearities are demonstrated. Keep the terms visibly separate in the manuscript.

2. Institutional history and as-of legal coding

The master legal timeline is a **table of policy instruments**, not a collection of timeless assertions. Each instrument has publication, effective and suspension/replacement times; scope, licensing standard, affected technical classification, destinations and end uses; official-language canonical reference; and frozen source artifact/hash. The April 4, 2025 MOFCOM/GACC Announcement No.18 covered specified samarium-, gadolinium-, terbium-, dysprosium-, lutetium-, scandium- and yttrium-related materials and required export licensing [R31]. November 7, 2025 Announcement No.70 suspended separately numbered measures **55, 56, 57, 58, 61, and 62** until November 10, 2026 [R32]. The April No.18 is **not** on that suspension list; treat it as operative on the research freeze *unless* a different superseding law is identified. That conclusion is an explicit inference, not a blanket legal opinion. Nd and Pr are central to magnet recipes, but the April measure cannot be summarized as a blanket prohibition of all Nd/Pr exports.

The U.S. series begins with October 2022 advanced-computing/manufacturing restrictions, continues through October 2023 and 2024 refinements, the December 2024 controls of software, HBM and equipment, subsequent 2025 changes affecting foreign-owned fabs, and January 2026 advanced-chip licensing revisions [R26, R30]. The Netherlands' authorization requirement was expanded in April 2025, explicitly **not a categorical ban** [R28]. Japan amended equipment controls covering 23 high-performance categories with implementation from July 2023 [R29]. The legal codebook must respect that allied rules are *national actions* and do not necessarily carry the same destinations, criteria or extraterritorial jurisdiction as BIS instruments.

Minimum legal-event schema

Field	Type / rule	Why indispensable
`policy_id`, `instrument_id`, `version_id`	Immutable string keys	Keeps amendment/suspension separate from initial rule
`jurisdiction`, `issuer`, `official_language`	Controlled categorical	Allied controls are not interchangeable
`publication_date`, `effective_date`, `end_date`	ISO date, nullable	Legal timing versus public information
`info_public_date`, `retrieval_timestamp`, `sha256`	Point-in-time traceability	No-lookahead backtests
`technology_layer`, `product_code`, `legal_citation`	Many-to-many normalized mapping	Specific ECCN / Chinese dual-use / Dutch / Japanese codes
`destination`, `end_user`, `end_use`, `person_scope`	Scope fields	Authorization depends on transaction
`license_requirement`, `review_presumption`, `exemption`	Explicit coded values	Do not equate a license rule with prohibition
`status`	Proposed / in force / stayed / repealed / superseded	Correct historical treatment
`severity_ordinal`, `scope_coverage`, `confidence`	Separate measures	Severity score alone is not treatment intensity
`source_id`, `adjudication_id`	Foreign keys	Source audit and disputed readings

Pre-policy treatment exposure uses frozen **2021 or earlier** weights, never quantities that may have changed *because* of the restriction. Because ECCNs and HS classifications are not one-to-one, maintain a manual technical crosswalk with uncertainty flags and source evidence. A HS heading alone cannot show whether the imported machine or chip is at the relevant controlled technology frontier.

3. Rare-earth supply-chain accounting: objects, geography, and units

3.1 Supply scope and critical element bookkeeping

Fixed base basket: $E = \{\mathrm{Nd}, \mathrm{Pr}, \mathrm{Dy}, \mathrm{Tb}\}$. Preserve each element's identity through the physical network and publish element-level bottlenecks. A practical base geography B includes U.S. and agreed aligned consuming/processing jurisdictions; the membership table is a frozen input, not a verbal assertion that everything outside China is allied. Track **three distinct properties**: facility location, ownership/control (including ultimate parent), and feedstock origin. A processing facility in one country may require feedstock from another; Chinese-origin material processed in a friendly jurisdiction cannot be counted as wholly China-independent in a China-removal scenario. Alternate geographic definitions are published as sensitivities.

Stages: $s = 0$ mineral extraction/concentrate, $s = 1$ separated element-specific oxide, $s = 2$ converted metal/alloy or magnet powder, and $s = 3$ finished qualified magnet. An oxide is not the same physical object as a magnet. Raw native reported quantities include ore tonnes, mixed-REE oxide tonnes, contained Nd tonnes, mixed NdPr oxide tonnes, metal tonnes, alloy tonnes and finished magnet tonnes. Every source observation carries `native_unit` and `product_basis`; transformations occur only through a versioned recipe/conversion table with evidence and uncertainty. For example, an idealized Nd_2O_3 mass is not one-for-one Nd metal mass; conversion follows molecular weights and actual recovery. A

new stage does not create additional contained rare-earth element. A recycling feed counts only at its external entry point; internal returns must not be added again as exogenous supply.

Quarterly principal clock. Project statuses, shocks and output are evaluated at quarter end, but source events retain exact dates. Choose whether quarterly operating capacity means average quarter production or quarter-end run-rate; **headline feasible supply is quarter-total output** in tonnes of contained element, **not** an annual capacity divided ad hoc across studies. Store nameplate as reported per year and transform annual to a quarterly capacity ceiling using operating days/seasonality, such as $C^Q = C^A/4$ only under explicit uniform-calendar assumptions. Ramps then apply to attainable quarter-level capacity. Stockpiles are opening inventories with flow conservation and no free replenishment.

3.2 Project-stage capacity before physical routing

For a project-stage activity a, quarter t, let C_{at}^{native} denote rated throughput or output, $\bar{U}_{at} \in [0,1]$ maximum achievable utilization, T_a independently verified commercial commissioning quarter, $r_a > 0$ a ramp speed per quarter, and $\omega_{a,e}$ the proportion or conversion into contained element e. The *available stage activity bound* is

$$\bar{q}_{a,t} = \left(\frac{C_a^A}{4}\right) \bar{U}_{a,t} \left(1 - \exp[-r_a(t - T_a)_+]\right) \mathbf{1}\{t \geq T_a\},$$

where $(t - T_a)_+$ counts elapsed quarters. This makes activity zero before verified commissioning; an operational datum at quarter T_a may require a measured initial-ramp offset or a discrete observation rule if observed output is already positive. **Do not use this ramp formula mechanically for activities with an immediate step-change:** registered ramp classes include step, exponential, logistic and observed output trajectory, chosen by out-of-sample validation. For each facility at multiple stages, activities get different T_a, r_a and input recipes. If installed nameplate changes after expansion, version capacity rather than silently replacing the earlier history.

Useful discrete project states are A announced, F feasibility/permitting, I financing/financial investment decision, C construction, Q testing/commissioning, O operating, S suspended, X cancelled. Suspension is potentially reversible; cancellation may be absorbing absent an independently coded relaunch. FID is not commercial commissioning, and a company press release is not equal to independent confirmation of stable qualified output. Existing operating sites should be left-truncated or initialized in the risk state observed when reliable surveillance begins.

3.3 A network-flow production model rather than stage totals

Use a directed *multi-commodity process-and-transport network.* A commodity is a pair $k = (\text{product form}, e)$ where valid; for nonseparable alloy/magnet grades, keep a vector recipe rather than pretend every element can be independently manufactured. A physical site has one or more production activities a. Each unit of activity requires $b_{ak} \geq 0$ units of commodity k and generates $d_{ak} \geq 0$ output units of commodities. These coefficients incorporate grade-specific recovery yields and stoichiometric conversion. Joint-product processes that yield multiple elements share the **same** activity variable, avoiding the common mistake of running independent Nd, Pr, Dy and Tb capacity at full nameplate simultaneously.

Let $q_{a,t} \geq 0$ be activity throughput; $F_{h,k,t} \geq 0$ be commodity flow on admissible transport edge h; $Z_{v,k,t} \geq 0$ be end-of-quarter inventory; $M_{v,k,t}$ verifiable external imported supply (not already an edge); and $d_{g,t} \in [0, D_{g,t}]$ quantity of final magnet service grade g served to bloc demand. For each node v, commodity k, quarter t, impose

$$Z_{v,k,t} = Z_{v,k,t-1} + M_{v,k,t} + \sum_{h:h\to v} F_{h,k,t} + \sum_{a\in v} d_{ak} q_{a,t} - \sum_{h:v\to h} F_{h,k,t} - \sum_{a\in v} b_{ak} q_{a,t} - L_{v,k,t},$$

with nonnegative stocks, physically meaningful storage limits and $L_{v,k,t}$ final-use deliveries. Finished grade demand satisfies $0 \leq d_{g,t} \leq D_{g,t}$, L corresponding to magnet quantities or application-service units. Process capacity: $0 \leq q_{a,t} \leq \bar{q}_{a,t}$. Routing: $0 \leq F_{h,k,t} \leq \bar{F}_{h,k,t}$ and unavailable edges fixed to zero. Shared machine/electricity limits: $\sum_{a\in\mathcal{A}(v)} \phi_{a,r} q_{a,t} \leq R_{v,r,t}$ for resource r. A magnet grade with required element mass $\theta_{g,e}$ consumes $\theta_{g,e} d_{g,t}$ of that element, with appropriate other inputs, scrap/waste and recovery entered in activity matrices. Recipes may vary only within preregistered metallurgical feasibility bounds; arbitrary zero-Dy substitution is not assumed.

Principal linear program, single quarter:

$$\max_{q,F,Z,d} \sum_g w_g \frac{d_{g,t}}{D_{g,t}}, \qquad \text{subject to all node balances, activity capacities, trade edges, grades, and inventories.}$$

Predefine $w_g \geq 0$, $\sum_g w_g = 1$, using *frozen priority/service weights*. In a second lexicographic solve (fix primary maximum to within tolerance), minimize shipping and imported-input use to produce a deterministic preferred feasible allocation. Publish **also** the minimum fraction across strategically essential grades, $\min_{g\in G_*} d_{g,t}/D_{g,t}$, because weighted service can mask a single high-importance bottleneck. If all final services are substitutable across grades, record that technological assumption explicitly; otherwise keep separate demand bounds. For rolling forecasts, the multi-quarter optimization also constrains intertemporal inventory and cannot consume the same stockpile every quarter.

Solver acceptance: all commodities nonnegative; every node balance residual below a unit-scaled tolerance; no production above capacity; recovery never creates contained element; joint activity constraint never double counts mixed feed; row/column dimensional checks; predicted completed magnet grade is supported by upstream oxides, metals and alloy quantities. An LP infeasibility is a diagnostic; do not fill the gap by inserting unconstrained imported feed. Mixed-integer variables are warranted only for indivisible plant on/off decisions or batch qualification, and complexity must be measured before choosing them as the headline model.

3.4 Distinguish three resilience outcomes precisely

The drafts used the word **resilience** for three different objects. The master fixes:

1. **Uncapped potential coverage** $C^{\text{raw}}_{k,t} = K^{\text{feasible potential}}_{k,t}/D_{k,t}$ at a particular stage/product k. It can exceed one; it describes surplus potential. Do not impose the unit interval on it.
2. **China-free deliverability** $Y^{\text{CN0}}_t = \sum_g w_g d^{(\text{remove China})}_{g,t}/D_{g,t} \in [0,1]$, with all country-of-origin/control and routing exclusions explicitly implemented and demand fixed. This is the **primary game state mapping** $y_t = Y^{\text{CN0}}_t$ for the U.S.-aligned bloc.

3. **China-free-plus-one-outage robustness** $Y_t^{\mathrm{CN0,N-1}} = \min_{r \in \mathcal{R}_{\mathrm{exCN}}} \sum_g w_g d_{g,t}^{(-\mathrm{China},-r)} / D_{g,t} \in [0,1]$, with a *single* additional supplier r removed from **all** stages, edges and ownership links and the **entire LP re-solved**. This is the main operational redundancy **stress metric**, and an alternative game state in robustness analysis.

The phrase "China-free" does not mean "non-Chinese downstream plant regardless of upstream feedstock." Demand $D_{g,t}$ remains the same when supply is removed. Unlike a global IEA-style N-1 country-accounting convention that may remove both a country's supply and demand, the counterfactual here specifically asks whether *the importing bloc's demand* is serviceable after losing an external supplier. Geographical and ownership stress definitions must be shown side by side. A policy control scenario may license certain Chinese material instead of fully removing it; parameterize export approval fractions by product and time rather than treating controls as a deterministic full outage.

Definition of worst-one-supplier: for each country-group or corporate control group r, simultaneously deactivate its upstream and downstream assets and affiliated routing rights, then solve the alternative feasible flow problem. Never subtract the largest provider independently for every stage. If a supplier is only a transport/logistics provider, its affected edges are removed and nodes remain when appropriate. For uncertainty about ultimate ownership, compute adversarial upper/lower resilience scenarios; do not silently classify mixed-control sites.

3.5 Fully traceable synthetic capacity example

All figures in this paragraph are **toy assumptions**, not data. All amounts are *tonnes of contained NdPr element per quarter*, and the final product is defined in NdPr-contained-equivalent tonnes, so conversions are directly comparable. Supplier groups A and B have separation-oxide output ceilings 60 and 45, metal/alloy output ceilings 50 and 40, and finished magnets (contained-NdPr equivalent) ceilings 40 and 35. Oxide-to-alloy element recovery is 0.90; alloy-to-finished-magnet element recovery is 0.95. Assume inter-group routing is feasible and enough other magnet constituents, energy and manufacturing inputs exist. Bloc final demand is 70 tonnes contained-equivalent per quarter. No inventory or other supplier exists.

Scenario	Available separated element	Max alloy element after recovery and capacity	Max magnet-contained element	Deliverable service / demand
A and B functioning	105	$\min(90, 0.90 \times 105) = 90$	$\min(75, 0.95 \times 90) = 75$	$\min(75,70)/70 =$ **1.000**
Remove A from all stages	45	$\min(40, 0.90 \times 45) = 40$	$\min(35, 0.95 \times 40) = 35$	**0.500**
Remove B from all stages	60	$\min(50, 0.90 \times 60) = 50$	$\min(40, 0.95 \times 50) = 40$	**0.5714**

The N-1 robust served fraction is $\min(0.5, 0.5714) = 0.5$. The no-outage **raw final-capacity coverage** is $75/70 = 1.0714$, while delivered service fraction is capped at one. These are different, compatible facts. Stage capacity sums (105 + 90 + 75) would be a categorical double-counting error. This example intentionally uses identical contained-element units; the production repository must test richer oxide/metal/magnet conversions and joint Nd/Pr/Dy/Tb recipes. Note that the *actual LP* may choose routing optimally; this simplified example is a closed-form test fixture for its maximum output, not a substitute for optimization in the real network.

3.6 Data dictionary: project, activity and material-flow tables

Key variable(s)	Type / unit	Source hierarchy and as-of requirement	Validation and treatment of missing values
`project_id`, `facility_id`, `activity_id`	Persistent strings	Corporate filings, official permitting and commissioning	One activity per independently capacity-limited stage; entity resolution manual log
`site_country`, `owner_parent`, `control_group`, `origin_group`	ISO3 / entity IDs	Audited subsidiaries, official records, customs origin where available	Ambiguous beneficial control preserved as scenario
`stage`, `product_code`, `element`, `magnet_grade`	Versioned controlled categories	Technical company documents, chemistry specs	Never silently convert product form
`nameplate_native`, `capacity_basis`, `native_units`	t/year or native throughput	Audited annual report > regulatory filing > company IR	Missing nameplate remains missing with explicit capacity-interval prior
`oxide_to_element`, `composition`, `yield`	Dimensionless / interval	Technical source/specification, verified production evidence	Hierarchical interval/posterior; reject wrong unit conversions
`commission_q`, `verified_operating`, `target_commission`	Quarter / bool	Audited operations and government filings	Interval censor uncertain dates; never overwrite old targets
`status`, `stage_age`, `transition_date`, `suspension_relaunch`	Categorical/date	Archived dated milestone announcements	Dual source for subjective milestones; competing-risks interpretation
`ramp_speed`, `utilization`, `actual_output`	1/quarter, fraction, t/q	Filings and production data	Separate rated maximum from realized production
`input_node`, `output_node`, `flow_commodity`, `max_route`	IDs, t/q	Transport, offtake, licensing, technical disclosures	Missing route = **unknown**, not zero or infinite; use scenario bounds
`inventory_open`, `inventory_close`, `stock_age`	t contained element	Audited inventories where obtainable	Unknown treated with explicit bounds, not free supply
`demand_grade`, `magnet_recipe`, `strategic_weight`	t/q, elemental fractions	IEA demand/engineering/industry scenario	Lock grade weights before outcomes; report interval uncertainty
`announcement`, `FID`, `financing`, `subsidy`, `offtake`	dates, USD real, status	Government award, EDGAR, exchange announcement	Announced grants ≠ disbursed funds; signed ≠ aspirational MOU
`price_element`, `policy_event`, `spoi`, `geo_risk`, `tech_risk`	USD/kg, score, controls	Official price series/archived legal sources	Source and date alignment; do not claim opacity is identical to risk
`valid_date`, `info_public_date`, `retrieval_date`, `source_sha256`	ISO date, hash	Every observation	Exclude origin if information publication date not known

Archive eligibility. All projects with at least one verifiable public announcement of an extractive/separation/metallurgy/magnet capacity asset in the defined geography, and with usable project ownership/stage/date evidence, enter the **candidate universe**, including cancelled and delayed projects. Exclude purely speculative media mentions without a discoverable owner/site/capacity project identity from the statistical risk set but retain them in a *nonqualifying discovery log*. Do not remove projects ex post because they failed. Freeze eligibility rules and historical cohort definitions before fitting event histories. Expansion phases are separate capacity activities under the same project parent to avoid

counting both original and expansion aggregate nameplate twice. Confidential deals are a coverage limitation, not evidence they did not occur.

4. Project transitions, uncertainty, and rare-earth probabilistic forecasting

4.1 Multistate competing-risks estimation

Project j enters state a at quarter t. Let $p_{j,a\to b,t}$ be the probability of transition $a \to b$ during that quarter, conditional on being in state a at the beginning. The main model is a discrete-time cause-specific hazard built from a shared competing-risk structure, with duration since state entry and calendar-quarter baseline:

$$\lambda_{j,a\to b,t} = \exp\{\alpha_{ab}(\text{age}_{j,a,t}) + \zeta_{ab,t} + X'_{j,t-1}\beta_{ab} + u_{\text{country}} + v_{\text{stage}}\}.$$

A **coherent** joint-quarter categorical transition law is

$$p_{j,a\to b,t} = \{1 - \exp[-\Lambda_{ja,t}]\}\,\frac{\lambda_{j,a\to b,t}}{\Lambda_{ja,t}}, \quad p_{j,a\to a,t} = \exp[-\Lambda_{ja,t}], \qquad \Lambda_{ja,t} = \sum_{b\neq a} \lambda_{j,a\to b,t}.$$

This treats the λ's as piecewise-constant within-quarter cause-specific intensities; it ensures probabilities sum to one, unlike naively estimating many independent cloglog hazards and adding them. When only one transition is eligible, $\log[-\log(1-p)] = \log\lambda$, recovering the familiar complementary-log-log link. Do **not** permit impossible transitions (e.g., planned directly to commercial operation without evidence) unless retrospective reporting is represented as a separate *observation* model. Registered time-varying covariates include FID, committed financing, permit status, binding offtake, subsidy type, prices, input scarcity, equipment readiness, project age, revision/slippage, policy events, regulatory opacity and calendar shocks. Include country/stage random intercepts if sample permits. Correct for calendar clustering and common project group exposures.

Right censoring: a project announced in 2025 and still building at the 2026 freeze has not failed just because it has no observed output. **Interval censoring:** operation publicly confirmed in September 2024 after a December 2023 update may have commenced anywhere in between. **Delayed public reporting:** information about a March 2025 commissioning event published in June 2025 cannot be used in forecasts made in April 2025. Use the first independently public confirmation rather than retrofitting the true operating date to earlier vintages. **Cancellation/long suspension:** cancellation is a competing event; choose how long an indefinite suspension remains at risk using explicit rules. Fine–Gray may summarize cumulative incidence but its subdistribution hazard is not substituted mechanically as transition probabilities in the project simulator [R12–R13].

Large technical projects are a small, heterogeneous population. Model complexity is limited by *observed transitions*, not by number of rows after quarterly expansion. Predefine a small covariate baseline, shrinkage or hierarchical pooling, and a protocol for dropping unidentifiable categories. Do not manufacture precise delay parameters if only a handful of refineries or metallization plants ever reached commercial operation.

4.2 Sources of dependence and forecast uncertainty

Separate at least four uncertainty levels: Θ project-transition model and estimation uncertainty; Z_t common shocks (price, regulation, financing, technology, geopolitical block); T_{js}, r_{js}, U_{js} project-specific commissioning and ramp; and A, Y, θ physical network-routing, process yields, demand, recipes and inventories. Simulation draw d proceeds sequentially:

1. Draw admissible model parameters and shared scenario path $Z_t^{(d)}$, conditioned on information available at forecast origin o.
2. Simulate every project's next allowed state, including suspension and cancellation, conditional on common and idiosyncratic covariates; preserve ownership-stage correlations.
3. Draw stage-specific commissioning, output qualification, ramp and utilization; treat a downstream plant dependent on a shared upstream input as jointly exposed.
4. Draw physically plausible yields, elemental mix, inventories and demand by grade; reject inconsistent units/recipes and explicitly track data gaps.
5. Solve the joint flow network in every quarter and under each registered policy/outage scenario, including one-outage reruns.
6. Record raw surplus coverage, capped served fractions, bottleneck identity, missing-input diagnostics, and first-passage history.

The Monte Carlo **headline event** is

$$\tau_h^{\mathrm{CN0,N1}} = \inf\{q : \ Y_{q+i}^{\mathrm{CN0,N1}} \geq h \text{ for } i = 0, 1, 2, 3\},$$

with $h \in \{0.5, 0.75, 0.9, 1\}$. Define the **recognition date** as quarter $q+3$, the fourth confirmed quarter, not retrospectively represent q as public knowledge of future persistence. Reports must distinguish $P(\tau_h \leq T)$ from probability the four-quarter persistence has been *verified by* date T; the latter requires $q+3 \leq T$. The forecast state and horizon always include the chosen distinction. If the theoretical ODE's target 1 is asymptotically unattainable from below for finite constant effort with positive depreciation, that is a property of the ODE, **not** evidence the physical LP cannot produce full quarterly service. Never use the ODE to overwrite a project simulation's physical first passage.

Monte Carlo accuracy: for binary hitting probabilities require $\sqrt{\hat{p}(1-\hat{p})/N} \leq 0.005$ before stopping and account for effective sample size under correlated weighted draws. Report predictive 5/25/50/75/95% quantiles of completion dates including right-censored/no-crossing mass, not falsely finite medians. Report 50%, 80%, 95% uncertainty intervals for final physical throughput and bottleneck layer, and sensitivity to Chinese material present in allied plants. Baselines must include zero-new-build, announced-on-time, historical mean delay, linear trend, and aggregate logistic trend. If sophisticated model does not improve valid historical prediction, report that negative result.

4.3 Historical-vintage backtests and correct scoring windows

The data freeze is **2026-10-11**, and future end dates are *not* yet observed. An as-of forecast reconstructed at origin o may use only records with `info_public_date` $\leq o$. A revision published in 2025 about a 2021 quantity may inform a 2025-origin fit but **not** a 2022-origin fit. Freeze training labels, project universes,

covariates and model-selection decisions at each origin. The following endpoints are mechanically eligible **subject to the actual outcome's release lag**:

Origin	+2-year endpoint	+4-year endpoint	+6-year endpoint
2016-12-31	2018-12-31: potentially scoreable	2020-12-31: potentially scoreable	2022-12-31: potentially scoreable
2018-12-31	2020-12-31: potentially scoreable	2022-12-31: potentially scoreable	2024-12-31: potentially scoreable
2020-12-31	2022-12-31: potentially scoreable	2024-12-31: potentially scoreable	2026-12-31: **unobserved**
2022-12-31	2024-12-31: potentially scoreable	2026-12-31: **unobserved**	2028-12-31: **unobserved**
2024-12-31	2026-12-31: **unobserved**	2028-12-31: **unobserved**	2030-12-31: **unobserved**

A four-quarter sustained threshold observed through the endpoint may not be **confirmed** at the endpoint unless all four requisite quarters have elapsed. For an event defined by its *start date*, only confirm retrospectively if the final quarter has ended and the source has been published by freeze. Project-level hazard event counts usually permit more informative proper scoring than the handful of overlapping system-level rare-earth threshold realizations. Cluster/block-bootstrap by project parent and common calendar origin to avoid treating heavily overlapping forecasts as independent observations. Reliability diagrams on three or five system forecasts are descriptive at best; do **not** claim statistical calibration or tight confidence bands. Use Brier and log score for fixed-horizon events, CRPS and empirical interval coverage for continuous capacity, and time-dependent Brier/CIF calibration for commissioning risks [R17–R18]. Sample availability failures must be logged, not selectively discarded after seeing which forecasts were accurate.

4.4 The actual turning-point deliverables

The rare-earth chapter must produce: quarterly projects-at-risk inventory; achieved and anticipated capacity by *stage* with credible intervals; graph of feasible element flows, not sums; demand assumptions and scenario registry; China-free service; China-free+N1 service; worst-hit supplier and stage/element/grade bottleneck; forecast $P(\tau_h \leq T)$ at 2028Q4, 2031Q4, 2035Q4; historical-vintage forecast scores relative to baseline; and the distribution of required financing, commissioning and offtake milestones. Define "China's chokepoint ended" only as a scenario-specific *service adequacy* finding; model cannot infer all geopolitical leverage is eliminated by crossing one physical threshold.

5. Differential equations: viability threshold versus true bifurcation

5.1 A deliberately reduced-form physical resilience ODE

A useful aggregate approximation after the project model is

$$\dot{r} = \eta I(t)(1 - r) - \delta r, \qquad 0 \leq r \leq 1, \quad \eta > 0, \quad I \geq 0, \quad \delta \geq 0.$$

Here r is a **normalized, bounded adaptation proxy**, not the raw K/D ratio and not guaranteed to equal the network LP service fraction at every date. Model validation must compare the proxy to the latter: if residuals show systematic bottleneck shifts, use stage-specific states or discard the ODE as a physical forecast. For constant I, with $A = \eta I$ and $\lambda = A + \delta$,

$$r^* = \frac{A}{A+\delta}, \qquad r(t) = r^* + [r(0) - r^*]e^{-\lambda t}, \qquad \frac{\partial \dot{r}}{\partial r} = -\lambda < 0.$$

The interval is invariant: $\dot{r}|_0 = A \geq 0$ and $\dot{r}|_1 = -\delta \leq 0$. For a threshold $0 < h < 1$, a finite, upward crossing from $r_0 < h$ requires strictly $r^* > h$, not merely equality. The sufficient and necessary constant-effort inequality is $I > \delta h/[\eta(1-h)]$; if equality holds, the system approaches h asymptotically and never crosses it from below. The crossing time is

$$T_h = \frac{1}{\eta I + \delta} \log\left(\frac{r^* - r_0}{r^* - h}\right), \qquad r_0 < h < r^*.$$

For $\delta > 0$ and finite I, $r^* < 1$: the exact target $h = 1$ is **not attainable** from below in finite time. Even with $\delta = 0$ and $I > 0$, $r(t) < 1$ at every finite t starting below one. This mathematical limit is why the physical network's 100% service event and a bounded reduced-form ODE must be maintained as **distinct** objects. Re-labeling the ODE threshold "tipping" does not establish multiple steady states or hysteresis.

5.2 A genuine economic tipping model recovered from Draft 0

Exploratory second model for adoption/completion propensity $z \in [0, 1]$:

$$\dot{z} = \kappa\{\Lambda(\theta + \beta z) - z\}, \qquad \Lambda(v) = \frac{1}{1 + e^{-v}}, \quad \kappa > 0.$$

Interpret z as a normalized propensity for *follow-on* projects to be completed or financed, **not** as tonnes of actually constructed capacity. θ includes fundamentals such as prices, availability of public support, technological feasibility and measured opacity. β is a scale-dependent strength of endogenous positive feedback: existing network learning, qualified skilled labor, counterparties, offtakes, finance confidence or shared processing infrastructure may make later investments more likely. Under these normalizations, fixed points obey $z = \Lambda(\theta + \beta z)$, equivalently $\theta = \log[z/(1-z)] - \beta z$.

The fixed-point curve has stationary points when $1/[z(1-z)] - \beta = 0$. Since $z(1-z) \leq 1/4$, **three fixed points are possible only when** $\beta > 4$. For $\beta > 4$, fold coordinates are

$$z_\pm = \tfrac{1}{2}\left(1 \pm \sqrt{1 - 4/\beta}\right), \qquad \theta_\pm = \log\left(\frac{z_\pm}{1 - z_\pm}\right) - \beta z_\pm.$$

The two outer fixed points are stable when $\beta z(1-z) < 1$; the middle point is unstable when $\beta z(1-z) > 1$. The exact θ interval supporting three fixed points is between the two fold values (ordered numerically). Hysteresis requires θ to change through the folds *without other structural rules invalidating the model*. Example: $\beta = 5$ gives $z_- \approx 0.2764$, $z_+ \approx 0.7236$. This threshold **is a property of a**

specifically normalized logistic interaction, not a universal industrial tipping coefficient. More robust parameterizations have other critical values.

Empirical test plan. Predict later financing/commissioning transitions conditional on *previously established* regional/stage-relevant operating supply-network density and interaction with financing/support; report correlated regional shocks, spatial contagion, and endogenous project siting as alternative explanations. Fit a hierarchical discrete-choice/transition model with frozen exposure windows and as-of observed network density; simulate the implied aggregate $\Lambda(\theta + \beta z)$ only if its micro-to-macro assumptions are tenable. Estimate $P(\beta > 4 \mid D)$ or a confidence region; compare monotonic non-tipping alternatives and out-of-sample predictions. A large estimated positive β without adequately controlled location selection, state dependence, and common macro factors does **not** show causal tipping. If the estimated $\beta \leq 4$ or no robust feedback is present, conclude that forecast project buildout is gradual rather than hysteretic. This is a legitimate and publishable negative outcome.

5.3 Shock propagation does not automatically imply coordination tipping

The material LP already permits **physical cascading shortages**: losing oxide feed can propagate to alloy then magnet output through conservation constraints. Reserve a separate, optional *financial/expectational contagion* network for observed common financiers, technology licensors and offtake partners. For node i, a stylized threshold response is $v_{i,t+1} = \mathbf{1}\{a_i + \sum_j w_{ij} v_{j,t} \geq c_i\}$. This is a conceptual mechanism; it cannot be calibrated faithfully without a substantial directed exposure map. A local linear approximation $r_{t+1} = Ar_t + \varepsilon_t$ amplifies perturbations when the spectral radius $\rho(A) > 1$ in that linear system, but neither calculation makes a physical LP outage a behavioral contagion event. The rare-earth empirical study should not expand into an unrelated bank/financial crisis research paper merely because Draft 0 surveyed that literature. The key additional question is whether *the success of one real industrial project causally makes subsequent projects more viable.*

6. Semiconductor controls, foreign access, and indigenous capability

6.1 Define a target, not an abstract national 'technology level'

A well-defined semiconductor target is a *vector of uses/performance requirements and production steps.* Before measuring anything, register a strategic target architecture a: advanced-node logic with a specified production-volume, energy/performance, reliability, yield and memory/packaging envelope. Different targets (e.g., advanced AI compute, military aerospace, mature automotive controllers) require different technology stacks and may not have the same weakest link. Do not infer advanced-chip capabilities from aggregate integrated-circuit unit production or from patent counts. Avoid a single national "nm" measure spanning different foundries, architectures and yield rates.

Registered layer set $\mathcal{L}$: (1) design automation, verification and essential design/process intellectual property; (2) lithography and critical masks; (3) deposition, etch, ion implantation and annealing; (4) metrology/inspection, cleaning and process control; (5) qualified advanced foundry integration and commercially usable yield/throughput; (6) relevant advanced memory/HBM; and (7) advanced packaging, interconnect and integration. A target may require only a subset (e.g., HBM may not bottleneck all products); set $\mathcal{L}(a)$ separately and before observing outcome. The data must be granular enough to

distinguish indigenous production of *critical layers* from temporary foreign equipment already installed or parts sourced abroad.

For each ℓ, a, t, record distinct bounded states: $z_{\ell,a,t} \in [0,1]$, **demonstrated domestic capability** toward that fixed target; $f_{\ell,a,t} \in [0,1]$, foreign capability still accessible and usable under contemporaneous rules and commercial/logistical constraints; and $a_{\ell,a,t} \in [0,1]$, effective availability. The principal illustrative combination is

$$a_{\ell,a,t} = 1 - (1 - z_{\ell,a,t})(1 - f_{\ell,a,t}) = z_{\ell,a,t} + (1 - z_{\ell,a,t}) f_{\ell,a,t}.$$

The formula assumes foreign access can fill the portion of the target not achieved domestically. This is a **substitutability approximation**, not an engineering identity for every production step. Where domestic and foreign inputs are complementary rather than substitutes, estimate a target-specific production possibility function $a_{\ell,a,t} = H_\ell(z, f; \psi_\ell)$, potentially $\min\{z, f\}$ or a constrained recipe. Publish both and flag layers where the simple formula is unjustified. The strict bottleneck targets are

$$S_{a,t}^{\text{domestic}} = \min_{\ell \in \mathcal{L}(a)} z_{\ell,a,t}, \qquad S_{a,t}^{\text{effective}} = \min_{\ell \in \mathcal{L}(a)} a_{\ell,a,t}.$$

These serve two different questions: *what can China make indigenously?* and *what can the target production system obtain at present?* **Only domestic capability** is mapped to the slow Chinese adaptation game state x_t, because the immediate legal access shock should not be misidentified as delayed learning. A differentiable harmonic/soft-min is a computational sensitivity, not the descriptive standard. Define, for positive $z_\ell + \epsilon$, $S_q(z) = [\sum_\ell w_\ell (z_\ell + \epsilon)^{-q}]^{-1/q}$, and optionally remove the ϵ bias with a documented correction or use a smooth log-sum-exp approximation. Pre-register weights and q; don't retune them to fit policy outcomes. The literal min must remain reported.

6.2 Frozen legal-intensity map and exposure

Let $C_{r,\ell,a,t} \in \{0, 0.25, 0.50, 0.75, 1\}$ be an **ordinal** severity coding for restriction by jurisdiction r, layer ℓ, target/user a, quarter t, with anchors: 0 no special control; 0.25 documentation/limited permission; 0.50 individual application with meaningful possible approval; 0.75 broad authorization with restrictive review; 1 functional denial or near-prohibition for the specified transaction. Separately code item coverage and share of supplier-dependent target input; do not assert a 0.75 rule creates exactly 75% supply loss. Denote frozen pre-2022 dependence shares $w_{r\ell a}^0$ summing to one for relevant foreign supply. The dimension-reduced legal exposure is $C_{\ell at} = \sum_r w_{r\ell a}^0 C_{r\ell at}$. It is a legal treatment index whose operational meaning requires empirical validation against actual access. Record direct-denial rules, FDPR applicability, U.S.-person restrictions, national-rule exemptions, licenses, servicing restrictions and maintenance/parts separately.

Identification risk: rules target strategically advanced firms and products; simple treated-versus-untreated comparisons can reverse causality. Only within a well-defined overlapping firm/product risk set can staggered event-study comparisons support conditional causal interpretations. Anticipatory purchasing before controls means the effect at publication may differ from the effect at legal entry into force. Public reports of illegal transshipment, licenses or underreported shipments have severe selection and observability limits. Don't classify a high observed import series as proof the legal regime is ineffective without checking whether technology, vintage and end use were actually controlled.

6.3 Strong operational evidence anchors for z and f

Use a versioned **evidence adjudication** scale for domestic capability at every layer/target. Base anchors are 0 no independently verified domestic qualifying capability; 0.25 laboratory/prototype or unqualified isolated system; 0.50 qualified limited commercial operation below frontier or limited throughput; 0.75 repeatable substantial production approaching the registered target with material capacity constraints; 1 independent evidence of the specified target quality, repeatability and throughput. Only assign a score when a source supports the **specific target**; a 0.75 in one frontier year is not automatically 0.75 after changing requirements. Maintain fixed-target and moving-frontier versions, with the difference itself reflecting technological obsolescence.

Layer	Highest-value capability observations	Low-quality proxies (never sufficient alone)
EDA/design IP	Verified production tapeouts for registered class; functioning sign-off, verification and IP ecosystem	Software announcement; patent family count
Lithography	Qualified wafer layer critical-dimension capability, stable overlay and throughput at target	Installed tool count; lab resolution claim
Etch / deposition / implantation	Qualified share of *critical* process steps at target yield and repetition; reliability	General machinery shipments; import value
Metrology / inspection / clean	Qualified measurement/defect-control throughput to sustain process yield	Inspection-equipment import aggregate
Foundry integration	Repeatable commercial yield, target-node volume, qualification for specified chip	Announced nm label; gross wafers
HBM/advanced memory	Usable memory bandwidth, stacks, yields and compatible product qualification	Aggregate DRAM unit production
Advanced packaging	Qualified package scale, interconnect density, thermal/yield and production availability	Patent filings or demo package

For foreign access f, triangulate authorized shipments, licensing rules, supplier disclosures, installed equipment and maintenance, inventory drawdown, partner substitution, and evidenced ability to use imported systems. Treat confidential licenses and diversion as partially unobservable; report bounds. Negative evidence (no disclosed indigenous tool) means "unverified" rather than proof zero capability when public disclosure is systematically selective. Each rating record has `as_of`, `source_id`, `coder`, `confidence`, `target_version`, and a written evidence memo. Two coders adjudicate borderline functional scores.

6.4 Keep negative policy effects and physical state bounds compatible

Previous drafts simultaneously allowed $\theta_\ell < 0$ in a linear investment pipeline and assumed $p_\ell \geq 0$, which is invalid unless explicitly constrained. Fix this by distinguishing **signed policy effects on effort** from **nonnegative realized effort**. For the layer-level adaptation model use

$$\dot{p}_\ell = \kappa_\ell\{J_\ell(C_\ell, X_\ell) - p_\ell\}, \qquad \dot{z}_\ell = [g_\ell + \eta_\ell p_\ell](1 - z_\ell) - \delta_\ell z_\ell,$$

where $\kappa_\ell > 0$, $g_\ell \geq 0$, $\eta_\ell \geq 0$, $\delta_\ell \geq 0$, and the **realized nonnegative effort intensity** is

$$J_\ell(C, X) = J_{\min,\ell} + \text{softplus}(\alpha_\ell + \theta_\ell C + X'\beta_\ell), \qquad \text{softplus}(v) = \log(1 + e^v), \quad J_{\min,\ell} \geq 0.$$

Choose and document effort units and normalization (e.g., investment/R&D intensity relative to an ex ante baseline). θ_ℓ **is sign-unrestricted.** If negative, greater restrictions can *reduce* effort, but cannot make realized physical investment negative. With $p_\ell(0) \geq 0$, the pipeline stays nonnegative because its target is nonnegative. $z_\ell \in [0,1]$ is forward invariant: at zero, $\dot{z}_\ell = g_\ell + \eta_\ell p_\ell \geq 0$; at one, $\dot{z}_\ell = -\delta_\ell \leq 0$. The physical conversion efficiency η_ℓ and policy-on-effort elasticity θ_ℓ are **different parameters** and should not be conflated. If sanctions also impair learning directly (equipment service withheld, knowledge network disrupted), register a second nonnegative inhibitory channel in η or δ, with sign estimates informed by evidence. The softplus response and its elasticities are *models to estimate*, not empirical findings.

Alternative transparent parameterization: use a log-link $J_\ell = J_{0,\ell} \exp(\theta_\ell C + X'\beta)$ when $J_{0,\ell} > 0$; it has identical positivity and sign properties but stronger multiplicative assumptions. Compare out-of-sample, monotonicity/curvature and effect sizes before choosing. The game below uses a *normalized* policy-adaptation rate function, not unscaled dollars or patent counts.

6.5 Two-step empirical program

Step A: immediate access effect. Define firm-by-product/month and destination-layer panels from official customs/trade, firm financials, regulation/event mappings and supplier reports; outcome variables can include log(1+quantity), real unit value, inventory, non-coalition sourcing share, equipment service continuity, and well-documented legal license approvals. Use exposure frozen pre-policy and distinguish publication from implementation. The pre-registered estimation estimand is group-time ATT for first treatment $G_i = g$, $\text{ATT}(g,t) = E[Y_t(1) - Y_t(0) \mid G_i = g]$, under conditional parallel trends with matched or not-yet-treated controls, if the treatment design supports it. Estimate cohort-time effects with Callaway–Sant'Anna or an appropriate Sun–Abraham implementation, not a naive heterogeneous-treatment TWFE lead/lag alone [R14–R15]. Cluster by treatment assignment (firm parent/sector where appropriate), allow event clustering, and use wild/bootstrap or randomization checks if few treated clusters. Plot pretrends, leads, overlap and attrition. If exposure is continuous, consider dose-specific designs [R16] rather than interpreting an ordinal 0/0.25/… index as exact physical dosage.

Step B: delayed domestic adaptation. Outcomes are separately R&D spending, capex, training, quality-adjusted patent families, qualified tool substitution, process yields, and evidence-coded capability z. The effects may have different signs and lags; e.g. spending can rise as actual frontier capability stalls. Pre-register windows of 0–6, 6–12, 12–24 and >24 months, subject to realized observability. A positive effect on R&D is not a positive effect on z. Model dynamic capability levels using the bounded state law and a measurement equation for the coarse 0/0.25/… ratings with coder uncertainty. If capability data are too thin, do not pretend to identify η_ℓ structurally: report estimated effort responses, observed milestones, and scenario intervals for the conversion into functional capability.

Threats to causal interpretation: selection into Entity List treatment; treatment anticipation; simultaneous Chinese industrial subsidies; COVID/global cyclical equipment shortages; direct effect of U.S. rules on American supplier revenue and R&D; policy spillovers to ostensibly untreated Chinese firms; heterogeneous treatment timing; globally moving technology frontier; record-based selection; and non-equivalent HS/technology crosswalk. Mitigation requires contemporaneous controls, event-time placebos, technology negative-control groups, exposure perturbations, sensitivity to Chinese subsidy announcements, and candid limits on causal language.

6.6 Semiconductor forecast schedule (release-lag conditional)

Origin	+6 months	+12 months	+24 months
2022-12-31	2023-06-30 possible	2023-12-31 possible	2024-12-31 possible
2023-12-31	2024-06-30 possible	2024-12-31 possible	2025-12-31 possible
2024-12-31	2025-06-30 possible	2025-12-31 possible	2026-12-31 **unobserved**
2025-12-31	2026-06-30 possible	2026-12-31 **unobserved**	2027-12-31 **unobserved**

"Possible" means the end date is before the data freeze, not that every necessary datum was actually published by then. A machine-checkable availability mask must check **outcome public-release dates** and target-version matching before scoring. Because the post-2022 observation window is short, forecasts should be short-horizon and clearly separate event-study estimation from predictive capability scores. Uncertainty intervals for latent capability must incorporate coder and source uncertainty; binary licensing policy changes alone do not establish technology substitution.

6.7 Semiconductor data dictionary

Fields	Units/frequency	Source / extraction logic	Important limitation
`policy_id`, `eccn`, `national_control_code`, `jurisdiction`, `scope`, `review`, `FDPR`, `in_force`	rule/event-date	BIS EAR/Federal Register; METI official controls; Dutch government orders	English summaries are not always definitive law
`frozen_dependency_w`, `base_year`, `supplier`	normalized share	2021/pre-shock supplier/firm dependence	Exposure may be unobserved or proprietary
`layer`, `target_version`, `architecture`, `evidence_memo`	categories/text	frozen technical specification and product evidence	Moving frontier must be separately versioned
`z_capability`, `f_foreign_access`, `a_effective`	0–1 quarterly	source-adjudicated functional tests and shipment/licensing evidence	Proxy measurement, uncertainty and substitutability assumptions
`firm_parent`, `firm_id`, `entity_list_event`, `treatment_cohort`	ID/date	BIS listings + securities filings	Sanctions are selected, not random
`r_and_d`, `capex`, `government_support`, `inventory`	real currency, q/y	audited SEC/exchange filings and official grants	Cross-currency conversion; parent subsidiary duplication
`exports_value`, `imports_quantity`, `hs_version`, `crosswalk_confidence`	USD, kg/units, month	USITC DataWeb; UN Comtrade; customs	HS $\neq$ ECCN; shipment quantities often incomparable

Fields	Units/frequency	Source / extraction logic	Important limitation
`patent_family`, `citation_quality`, `ipc_cpc`, `priority_date`	count/quality, event	WIPO PATENTSCOPE and patent offices	Patents are an activity indicator, not proof of industrial mastery
`frontier_target`, `qualified_yield`, `qualified_volume`	tech metric / q	filings, independent technical reports where admissible	Sparse public verification
`valid_date`, `info_public_date`, `source_hash`, `coder`, `confidence`	date/hash	immutable source archive	Without as-of date, not admissible to hindcast

7. Strategic Policy Opacity Index: measurement rather than imputed intent

7.1 Separate observability from the unobserved strategic motive

Woods [R03] explores a dynamic signaling mechanism in which Chinese information policy affects beliefs and diversification. The present study must not turn that theory into a presupposition that every missing statistic was withheld deliberately. **SPOI measures the objective information conditions facing an external investor**: timeliness, completeness and precision of relevant public information, licensing transparency, legal lead time, unexplained material policy reversals and regulatory consistency. It does **not** identify a policymaker's intent to deceive. Technological unknowns T_t and generic geopolitical/market volatility G_t are separate controls, and technology-related missingness without an institutional disclosure obligation is not automatically "strategic opacity." Build jurisdiction $\times$ policy-sector daily event panels; the primary empirical score is China/rare-earth-related policy opacity, with the U.S./semiconductor version a separately coded comparison.

7.2 Locked ten-indicator codebook and meaning of 0–4

Higher ordinal values mean **less observability**, anchored by prespecified thresholds. If an institutional obligation has no objective baseline, score *not applicable* (NA with reason), not 4. A score of 0 is affirmative measured timeliness/precision, not a guess that no problem exists.

Code	Measurement universe / anchor	0	1	2	3	4
O1	Mandatory scheduled disclosure lateness	by due date	1–30 d	31–90 d	91–180 d	>180 d / absent next reporting cycle
O2	Fraction of required fields missing from a scheduled release	≤5%	>5–15%	>15–30%	>30–50%	>50%
O3	Core export-license statistics available (applications/approvals/denials/pending/processing time), **only if expected**	≥90%	70–<90%	50–<70%	25–<50%	<25%
O4	Public-policy announcement-to-effective lead time	≥90 d	60–89 d	30–59 d	7–29 d	<7 d / retroactive
O5	Fraction of 4 legal scope fields explicit: product, target/use, procedure, review criteria	4/4	3/4	2/4	1/4	0/4
O6	Lead time before *material adverse reversal*	none or ≥90 d	60–89 d	30–59 d	7–29 d	<7 d / retroactive
O7	Still unresolved official-source inconsistencies (classified by materiality)	none	1 minor	1 material / 2 minor	2 material	≥3 material / material no resolution
O8	Fraction of *scheduled/obligatory* releases entirely absent	0%	>0–10%	>10–25%	>25–50%	>50%
O9	Share of reported throughput/capacity publicly classifiable by stage/product	≥90%	70–<90%	50–<70%	25–<50%	<25%
O10	Count of **material** legal amendments in prior rolling 90 days	0	1	2	3	≥4

Denominators are indispensable. O1, O2 and O8 rely on a dated release calendar with a documented duty or announced regular publication; O3 relies on a documented statistical series or duty to disclose. O9's total consists of a frozen universe of officially documented capacity/output records, not unobservable plants inferred by the researcher. If the expected denominator cannot be established, mark NA. O4/O6 measure **actual public policy-event lead times**, not volatility in news coverage. O7 requires written materiality rules; two coders cannot improvise "contradiction" after seeing project outcomes. O10 excludes purely typographical corrections. O1 and O8 overlap but measure different degrees of lateness/completeness: explicitly assess redundancy in construct validity tests.

7.3 Deterministic daily update and no-event rules

The earlier drafts left event scores undefined in quarters with no new announcement. This edition fixes the daily algorithm:

- **Scheduled disclosure stocks (O1, O2, O3, O8, O9):** calculate from the set of *active known obligations* at day d. O1 accumulates days overdue from the original due date. After release, retain its most recent disclosed completeness and lateness score only until the next scheduled release or a declared report-observation window ends. Define the principal reporting window as current active period, with all due series separately observed; if no required series exists for an indicator, return NA, not an imputed zero. O8 is a rolling 365-day share of scheduled releases entirely missed by each due-plus-grace checkpoint (grace 30 days principal; 0/90 sensitivity). O9 is computed from the contemporaneously public stage-granular series, not from ex post actual production.
- **New-policy shock O4:** on the **first public release** of a substantive policy with a known effective date, compute calendar-day lead time $\max(0, \text{effective} - \text{publication})$, with retroactivity placed in 4. The daily policy-shock score equals the worst still-active event whose public-release date lies in the previous 90 days. If none, **O4=0**, interpreted only as *no recent lead-time shock*, not as a claim every latent policy is transparent. Sensitivity windows: 30, 180 days, plus exponential decay. A future effective date never makes information retroactively public.
- **Current legal rule O5:** the scope clarity score of currently effective policy instruments persists until an instrument is clarified, expires or is superseded. If multiple simultaneously active rules exist, compute the weighted worst or exposure-weighted mean using **frozen pre-exposure weights** and publish the aggregation selected in advance; report individual instrument scores alongside the mean. No rule with no public record is not necessarily 0; document the coverage universe and score NA if it cannot be determined.
- **Material reversals O6:** on public announcement of an adverse reversal, compute the warning gap from publication to the change's effective date, and keep the event's severity for 90 days from *the later of publication and effective date* (with no retroactive leakage). Without a qualifying reversal in the window **O6=0**. A reversal quietly anticipated in markets but not publicly announced does not earn an invented warning date.
- **Unresolved conflicts O7:** material official contradictions persist *only while unresolved* and for a preregistered 90-day post-resolution impact window; code the originating evidence. No known conflict after an actual adjudication process means O7=0; missing search coverage means NA.
- **Rolling churn O10:** count independent material amendments with public-effective status in $(d - 90, d]$, de-duplicating amendments across agencies when one instrument is substantively one intervention. No events means 0.

For each day compute pillar scores only from observed indicators:

$$D_d = \frac{1}{4}\,\text{mean}(O1, O2, O3, O8, O9), \quad P_d = \frac{1}{4}\,\text{mean}(O4, O6, O10), \quad L_d = \frac{1}{4}\,\text{mean}(O5, O7).$$

The principal daily index is $\text{SPOI}_d = 100(0.40D_d + 0.35P_d + 0.25L_d)$. A day is valid only if $\geq$8 of 10 indicators observed, including $\geq$3 of 5 disclosure indicators, $\geq$2 of 3 predictability indicators and $\geq$1 of 2 legal clarity indicators. **Daily NA is not silently redistributed across pillar weights:** normalize each pillar by its observed denominator if valid and record coverage. The principal quarter score is the average across *valid calendar days*, requiring at least 80% of days in the quarter to qualify, and reporting coverage as a separate outcome. Also report quarter maximum as event-shock intensity, number of legal events and indicator counts. If quarterly coverage is <80%, return missing headline SPOI rather than LOCF or midpoint imputation. Regression models should use lagged SPOI_{q-1} to reduce mechanical simultaneity with project transitions, unless a publication-date event-study estimand requires a daily contemporaneous event.

7.4 Two full synthetic scores, reliability and missingness

Illustrative opaque-quarter *daily constant vector* $(O1, \ldots, O10) = (1, 2, 4, 4, 1, 3, 1, 2, 2, 3)$. The normalized pillar scores are $D = (1 + 2 + 4 + 2 + 2)/(5 \times 4) = 0.55$, $P = (4 + 3 + 3)/(3 \times 4) = 0.833333$, $L = (1 + 1)/(2 \times 4) = 0.25$; therefore $\text{SPOI} = 100(0.40 \times 0.55 + 0.35 \times 0.833333 + 0.25 \times 0.25) = 57.4167$. If this exact vector persists every day, the quarterly mean is 57.4167; this is **not a coded historical quarter**.

Illustrative transparent-quarter vector $(0, 1, 2, 1, 0, 1, 0, 0, 1, 1)$. Then $D = (0{+}1{+}2{+}0{+}1)/(5{\times}4) = 0.20$, $P = (1{+}1{+}1)/(3{\times}4) = 0.25$, $L = (0{+}0)/(2{\times}4) = 0$; index $= 100(0.40{\times}0.20{+}0.35{\times}0.25) = 16.75$. The labels only describe the *relative synthetic scores*; they make no claim about policy intent.

Reliability: double code O5, O6, O7 and a stratified sample of every other item; coders are blind to eventual project performance. Before full coding, preregister a target weighted Cohen $\kappa \geq 0.75$ for subjective categories and publish raw agreement and confidence intervals. If κ fails, revise the *codebook*, document the revision, independently recode, and prevent overfitting code definitions to investment outcomes. When statistical staff cannot access a scheduled source or determine an obligation, tag researcher-missing with reason; do not transform into highest opacity. Report alternative weighting and factor model separately. Validate against predictable dimensions (published license statistics, objective warning windows) and negative controls (irrelevant policy sectors); control for common calendar risk and source digitization trends.

7.5 Estimation of opacity drag and 'overreach'

Let $o_t = \text{SPOI}_t/100$, S_{jt} contractually documented government de-risking support, and T_t, G_t independent technological uncertainty and geopolitical/market uncertainty. In a registered project transition cause-specific intensity include

$$\log \lambda_{j,a\to b,t} = \alpha_{ab}(\text{age}) + \beta_1 o_{t-1} + \beta_2 o_{t-1}^2 + \beta_S S_{jt} + \beta_{OS} o_{t-1} S_{jt} + \beta_T T_t + \beta_G G_t + X_{jt}'\gamma.$$

For commissioning/progression, negative β_1 and positive β_2 may imply an "overreach" reversal at $o^* = -\beta_1/(2\beta_2)$; do not report a meaningful turning point unless $\beta_2 > 0$, o^* lies within the **observed**

support (ideally 5th–95th percentile), uncertainty bands exclude key opposite patterns, and alternative nonparametric shapes/missingness rules do not overturn it. Government support is not randomly assigned; the interaction is descriptive/conditional without a credible instrument or natural experiment. Stratify decisions by financing initiation versus successful commissioning: opacity could **increase hedging announcements while decreasing physical completion**, and a single hazard coefficient would obscure that distinction.

A theoretical investment heuristic $I(O,G) = I_0 + \gamma G - aO + bO^2 + hOG$ is useful for deriving hypotheses but can be negative. For physical simulations use either $I = \max(0, I_0 + \gamma G - aO + bO^2 + hOG)$ and acknowledge its kink or $I = \text{softplus}(I_0 + \gamma G - aO + bO^2 + hOG)$, with observation-scaled covariates. A positive fitted quadratic is insufficient to prove **strategic nondisclosure**; the SPOI is observability, and identification of intent needs separate credible archival evidence or a partially observed signaling model. Woods' illustrative numerical probabilities must never be presented as actual estimated resilience forecasts.

8. Reciprocal dynamic game: coherent, bounded and empirically anchored

8.1 Meaning of every variable and time unit

The empirical problems are not mirror images, but the final reciprocal game should be as small as possible while retaining the strategic feedback:

- $u(t) \in [0,1]$: standardized U.S.-led semiconductor restriction **policy intensity**, anchored to a frozen policy basket and observed U.S./allied legal severity; one unit is not "100% of all Chinese chip imports prohibited."
- $c(t) \in [0,1]$: standardized Chinese restriction intensity on the defined magnet-REE product/service basket, allowing licensing, partial approval and scope changes.
- $x(t) \in [0,1]$: China's slow-moving, domestically produced *functional semiconductor substitutability* for the registered stack, anchored to $x_t^{\text{obs}} = \min_{\ell \in \mathcal{L}(a)} z_{\ell,a,t}$ with measurement uncertainty and a differentiable alternative.
- $y(t) \in [0,1]$: U.S.-aligned *China-free magnet-service* resilience, anchored to the LP-delivered share Y_t^{CN0}; the stronger $Y_t^{\text{CN0,N1}}$ is a separately estimated robustness version.
- $m(t), n(t) \in [0,1]$: dimensionless delayed policy-exposure/adaptation pipelines caused by u, c, respectively; **neither is physical dollar investment** and both should not be fitted as observed production.

Continuous game time t is in **years**; quarterly empirical data are mapped via $\Delta t = 0.25$ years. All g, δ, κ, ρ are per year; β parameters below are dimensionless per unit normalized pipeline. Normalized payoff units are discounted utility or chosen monetary equivalents per year. All numerical values later specified are hypothetical examples unless explicitly backed by an empirical estimate.

8.2 Fixing the inherited mathematical inconsistency with sign-free responses

The common form in Draft 5, $\dot{x} = g_C + b_C m - \delta_C x$, is **not** constrained to $[0,1]$: at $x = 1$, its derivative is $g_C + b_C m - \delta_C$, which may be positive. Draft 5's fixed-policy state equilibrium was mistakenly taken from

a **different saturating equation**. The master chooses one coherent law, with a nonnegative growth rate even when sanctions suppress domestic adaptation:

$$\begin{aligned} \dot{m} &= \kappa_C(u-m), & \dot{n} &= \kappa_U(c-n), \\ \dot{x} &= G_C(m)(1-x) - \delta_C x, & \dot{y} &= G_U(n)(1-y) - \delta_U y, \\ G_C(m) &= g_C \exp(\beta_C m), & G_U(n) &= g_U \exp(\beta_U n). \end{aligned}$$

Primitive restrictions: $\kappa_C, \kappa_U > 0$, $g_C, g_U > 0$, $\delta_C, \delta_U \geq 0$; $\beta_C, \beta_U \in \mathbb{R}$ are *sign-unrestricted*. A negative β_C means U.S. controls reduce the effective growth/learning rate of Chinese domestic capability (relative to the baseline), without asserting that technical effort becomes negative. Positive β_C means controls accelerate it. The main game still allows a zero response $\beta_C = 0$. A negative β_U similarly means Chinese restrictions **impede** allied diversification rather than induce it. In applications where a linear growth function is empirically preferred, $G_C(m) = g_C + b_C m$ may be used **only with explicitly tested** $g_C + \min(0, b_C) \geq 0$ and analogously for the other player. Do not splice that linear function into an HJB derived for an exponential function without recomputing derivatives and steady states.

Invariance proof. At $m = 0$, $\dot{m} = \kappa_C u \geq 0$, and at $m = 1$, $\dot{m} = \kappa_C(u-1) \leq 0$. Same for n. At $x = 0$, $\dot{x} = g_C e^{\beta_C m} > 0$; at $x = 1$, $\dot{x} = -\delta_C \leq 0$. Same for y. The vector field is locally Lipschitz on the compact hypercube; by the inward-pointing boundary condition, $[0,1]^4$ is forward invariant for admissible measurable policies $u, c \in [0,1]$. The game cannot generate negative measured capability or above-unity resilience by integration if the numerical method preserves positivity and the solver diagnostics test it. Finite differences in a value-function solver must also enforce state-constrained, inward-looking stencils rather than arbitrary zero-value Dirichlet conditions.

Fixed controls: if u, c are constant, $m^* = u$, $n^* = c$ and

$$x^*(u) = \frac{g_C e^{\beta_C u}}{g_C e^{\beta_C u} + \delta_C}, \qquad y^*(c) = \frac{g_U e^{\beta_U c}}{g_U e^{\beta_U c} + \delta_U}.$$

The fixed-policy Jacobian is block triangular (after ordering pipelines before states) with eigenvalues $-\kappa_C, -\kappa_U, -[g_C e^{\beta_C u} + \delta_C], -[g_U e^{\beta_U c} + \delta_U]$, strictly negative for $g_C, g_U > 0$. This proves **asymptotic stability under constant policy**; it says nothing by itself about the stability of equilibrium *feedback* policies $u = \sigma_U(s), c = \sigma_C(s)$, whose full Jacobian must include $\partial\sigma/\partial s$. The time constant is an adjustment half-life $\log 2/[G_C(u) + \delta_C]$, not necessarily the literal half-life of remaining U.S. coercive leverage if baseline adaptation and depreciation alter the limiting vulnerability.

Model adequacy and falsification. Because a strict minimum across semiconductor layers can switch its binding layer abruptly and a physical network can open a discrete large plant, the projected empirical x_t, y_t need not follow a smooth four-state ODE exactly. Fit the reduced-form law as an approximation to annualized changes, test leave-out and regime-shift errors, and compare a stage/layer-augmented state-space model. If approximation errors systematically dominate the hypothesized coercive effect, abandon quantitative point predictions from the four-state game; retain it as a transparent scenario theory. A low-state Markov approximation is defensible only with explicit fit and predictive error accounting.

8.3 Discounted payoffs and what can realistically be identified

Keep the payoff structure from the mature drafts while maintaining sign-free complementarity:

$$\pi_U(s,u,c) = A_U u(1-x) - D_U c(1-y) - \tfrac{1}{2}k_U u^2 - \tfrac{1}{2}h_U n^2 + B_U y + q_U uc,$$
$$\pi_C(s,u,c) = A_C c(1-y) - D_C u(1-x) - \tfrac{1}{2}k_C c^2 - \tfrac{1}{2}h_C m^2 + B_C x + q_C uc.$$

Here A_i is marginal immediate strategic value from exploiting the opponent's current vulnerability; D_i is exposure to the opponent's coercion; $k_i > 0$ is convex marginal policy cost; $h_i \geq 0$ a cost of adaptation mobilization borne by the vulnerable side; $B_i \geq 0$ benefit from own resilience; $q_i \in \mathbb{R}$ net strategic coupling (retaliatory complementarity if positive; cost/substitution if negative). A positive q_i is **not** a structural theorem that retaliation always occurs. Each state $s = (x, y, m, n)$ has an objective

$$V_i(s) = E\left[\int_0^\infty e^{-\rho_i t}\pi_i(s(t), u(t), c(t))\,dt \;\middle|\; s(0) = s\right], \quad \rho_i > 0.$$

The expectation is redundant in the deterministic base game but prepares for stochastic versions. Compact state/actions and continuous bounded payoffs guarantee finite discounted objective, but **do not alone prove global pure stationary feedback MPE existence or uniqueness**. It is intellectually honest to normalize one payoff scale such as $k_U = 1$ in scenario analysis because **actions alone cannot identify the absolute cardinal scale** of strategic preferences; payoff parameters from sparse policy episodes are likely weakly or not identified. Avoid claiming the effect of a 2010 rare-earth episode recovers China's strategic cost curvature in a modern two-control game. Country strategic payoffs depend on more than observed shipping quantities.

8.4 Coupled stationary HJB and exact statewise constrained Nash

For $s = (x, y, m, n)$, let $f(s, u, c)$ be the four transition derivatives above. A stationary Markov-perfect candidate comprises differentiable/viscosity-appropriate value functions V_U, V_C and feedback controls σ_U, σ_C satisfying

$$\rho_U V_U(s) = \max_{u\in[0,1]}\{\pi_U(s, u, \sigma_C(s)) + \nabla V_U(s) \cdot f(s, u, \sigma_C(s))\},$$

$$\rho_C V_C(s) = \max_{c\in[0,1]}\{\pi_C(s, \sigma_U(s), c) + \nabla V_C(s) \cdot f(s, \sigma_U(s), c)\}.$$

These are **feedback** HJB equations. Open-loop Nash strategies chosen as functions of time or initial conditions are a different equilibrium concept and cannot be labeled the same solution. Because the policy control u enters the Markov dynamics directly only through $\dot{m} = \kappa_C(u - m)$, and c only through $\dot{n} = \kappa_U(c - n)$, the HJB first-order conditions, conditional on fixed continuation gradients, are **independent of the exponential functional form of** G_C, G_U:

$$M_U(s) = A_U(1-x) + \kappa_C \partial_m V_U(s), \qquad M_C(s) = A_C(1-y) + \kappa_U \partial_n V_C(s).$$

Thus the simultaneous exact best responses are

$$\boxed{\mathrm{BR}_U(c) = \Pi_{[0,1]}\left(\frac{M_U + q_U c}{k_U}\right), \qquad \mathrm{BR}_C(u) = \Pi_{[0,1]}\left(\frac{M_C + q_C u}{k_C}\right).}$$

Here projection truncates **each conditional best response**, not the pair of unconstrained simultaneous first-order solutions. If both controls are interior, $k_U u - q_U c = M_U$, $-q_C u + k_C c = M_C$, determinant $\Delta = k_U k_C - q_U q_C$ and

$$u_I = \frac{k_C M_U + q_U M_C}{\Delta}, \qquad c_I = \frac{k_U M_C + q_C M_U}{\Delta}, \quad \Delta \neq 0.$$

This pair is an interior candidate **only** if $0 < u_I < 1$ and $0 < c_I < 1$. A nonzero Δ is merely algebraic invertibility of the unconstrained linear equations, not proof that a unique bounded Nash equilibrium or a unique global dynamic MPE exists.

All nine KKT active-set regimes

An action pair is a statewise Nash candidate if both players' concave action Hamiltonians satisfy their proper boundary inequalities. Exactly one candidate may fail or multiple may coexist on knife-edge parameters; record all matches.

U regime	C regime	Candidate pair	Joint KKT feasibility
0	0	$(0, 0)$	$M_U \leq 0;\ M_C \leq 0$
1	0	$(1, 0)$	$M_U \geq k_U;\ M_C + q_C \leq 0$
0	1	$(0, 1)$	$M_U + q_U \leq 0;\ M_C \geq k_C$
1	1	$(1, 1)$	$M_U + q_U \geq k_U;\ M_C + q_C \geq k_C$
interior	0	$(M_U/k_U, 0)$	$0 < M_U < k_U;\ M_C + q_C u \leq 0$
interior	1	$((M_U + q_U)/k_U, 1)$	$0 < M_U + q_U < k_U;\ M_C + q_C u \geq k_C$
0	interior	$(0, M_C/k_C)$	$0 < M_C < k_C;\ M_U + q_U c \leq 0$
1	interior	$(1, (M_C + q_C)/k_C)$	$0 < M_C + q_C < k_C;\ M_U + q_U c \geq k_U$
interior	interior	(u_I, c_I)	$\Delta \neq 0;\ 0 < u_I, c_I < 1$

This table is the implementation contract for `solve_active_set`; at equality the solver must allow neighboring regimes to share an identical equilibrium without double counting it. Verify numerically that $|u - \mathrm{BR}_U(c)|$ and $|c - \mathrm{BR}_C(u)|$ fall below tolerance, and if desired compute one-sided payoff deviation maxima. **Counterexample to naive pair clipping:** take $k_U = k_C = 1$, $q_U = q_C = 0.8$, $M_U = 1.2$, $M_C = 0.1$. The unconstrained pair is approximately $(3.5556, 2.9444)$, whose independent clipping gives $(1, 1)$. But China's best response to $u = 1$ is $c = 0.9$: therefore $(1, 1)$ is **not Nash**; $(1, 0.9)$ is.

For fixed value gradients, the continuous map $T(u, c) = (\mathrm{BR}_U(c), \mathrm{BR}_C(u))$ sends compact convex $[0, 1]^2$ into itself, so Brouwer gives **at least one statewise action fixed point**. A sufficient condition for uniqueness is $|q_U q_C|/(k_U k_C) < 1$ from contraction of the two-step composed best-response map; the raw simultaneous map may use an appropriate weighted norm for contraction. Local stability under a hypothetical continuous-time *unconstrained* policy adjustment $\dot{u} = \lambda_U[(M_U + q_U c)/k_U - u]$, $\dot{c} = \lambda_C[(M_C + q_C u)/k_C - c]$ requires $q_U q_C < k_U k_C$, which differs from the absolute-product contraction condition. Neither result automatically settles existence, uniqueness or stability of the entire coupled HJB system.

8.5 Central analytical proposition: constant-control benchmark only

For a deliberately one-sided **constant restriction** $u \in [0,1]$, let $\dot{x} = bu(1-x)$, $b > 0$, $v(t) = 1 - x(t)$, $v_0 = 1 - x_0$, immediate coercive payoff $\pi = Auv - ku^2/2$, $A, k, \rho > 0$. The exact **constant-control objective** is

$$J(u) = \frac{Auv_0}{\rho + bu} - \frac{ku^2}{2\rho}.$$

The first derivative is $J'(u) = Av_0\rho/(\rho + bu)^2 - ku/\rho$; an interior constant-policy optimum solves

$$\boxed{ku(\rho + bu)^2 = Av_0\rho^2.}$$

The left side increases strictly in $u \geq 0$, ensuring a unique nonnegative root. With $Av_0 > 0$, $J'(0) = Av_0/\rho > 0$, so optimum is not zero. If the unconstrained root is < 1, it is the optimum; otherwise the optimum is 1. A myopic decision optimizes $Auv_0 - ku^2/2$, giving $u_{\text{myopic}} = \Pi_{[0,1]}(Av_0/k)$. At the forward-looking interior optimum, $ku = Av_0[\rho/(\rho + bu)]^2 < Av_0$, so $u_{\text{forward}} < u_{\text{myopic}}$ when the relevant myopic comparator is interior, and no stronger claim than weak inequality is made under saturated boundaries. Holding the other parameters fixed, the interior optimum rises with A, v_0, ρ and falls with b, k; parameters entering the bounds can make derivative zero in boundary regimes. This is a mathematically valid example of **self-eroding leverage**, but **does not** prove a general two-player MPE theorem or solve the unrestricted time-varying one-player problem.

Recomputable toy benchmark: $A = v_0 = k = \rho = b = 1$ produces $u(1+u)^2 = 1$, with unique interior root approximately 0.465571, less than the myopic bound of one. The exact residual of that claimed root must be tested in the reference code. Numerical claims from the 11-page summary that purported to solve full feedback laws without explicit payoffs are rejected.

8.6 HJB numerical method, algorithm and admissibility diagnostics

A candidate pure feedback MPE is obtained iteratively, not just by solving nine algebraic regimes at one state. A reproducible deterministic base algorithm is:

1. **Primitive registry.** Freeze $g, \beta, \delta, \kappa, \rho, A, D, k, h, B, q$ with estimation/calibration/scenario labels. Reject impossible input domains before solver start.
2. **Tensor state grid.** Start $N_x = N_y = N_m = N_n = 17$ (83,521 nodes), then 21 (194,481 nodes) and 31 (923,521 nodes) **only if memory/time feasibility demonstrated**; record sparse matrix sizes and measured runtime/memory, not assume a workstation handles a 31^4 nonlinear solve cheaply.
3. **Initial policy pairs.** Start from zero/zero, one/one, one/zero, zero/one, midpoint/midpoint and seeded random smooth policies. A single convergence path cannot establish uniqueness.
4. **Policy evaluation.** Holding σ_U, σ_C fixed, solve two separate sparse discounted linear equations $(\rho_i I - \mathcal{L}^\sigma)V_i = \pi_i^\sigma$. The generator $\mathcal{L}^\sigma$ uses dimension-wise **upwind** first differences: forward derivative when state drift is positive, backward when negative, state-constrained inward stencils at endpoints. The diagonal discount term ensures a proper monotone discounted discretization under admissibility.
5. **Policy improvement.** Derive $V_{U,m}$ and $V_{C,n}$ from grid-consistent one-sided differences; compute M_U, M_C, enumerate all nine KKT regimes at each grid point, record every distinct candidate, and update σ with damping/line search if the joint step oscillates. Re-evaluate until fixed point.

6. **Residual.** Define $\mathcal{R}_i = \rho_i V_i - \{\pi_i(s,\sigma) + \mathcal{L}^\sigma V_i\}$ as the **discrete continuous-time HJB residual** (units utility/year), and separately compute each unilateral deviation advantage $\mathcal{G}_U = \max_{u\in[0,1]} H_U(s,u,\sigma_C;V_U) - H_U(s,\sigma_U,\sigma_C;V_U)$, analogously for China. Do not write an undefined $\|B(V)-V\|$ without defining B.
7. **Acceptance thresholds.** Scale numerical residual tolerances by the payoff magnitude, e.g. $\|\mathcal{R}_i\|_\infty \le 10^{-8}(1+\|\pi_i\|_\infty)$ and action KKT $\le 10^{-7}$ at the selected mesh **if precision permits**, plus grid-refinement changes and held-out-policy verification. A small residual on a coarse grid is not proof of the exact continuous solution.
8. **Verification and equilibrium multiplicity.** Starting from each initial pair, classify distinct policies by sup norm and economically relevant policy decisions; if no pure candidate converges, refine, change damping or use a finite-state/action approximation and compute a stationary **mixed Markov equilibrium** under suitable discounted-game assumptions. Fink [R20] provides the existence basis for finite discounted stochastic games in mixed strategies, not an existence proof for continuous pure MPE.
9. **Closed-loop dynamics.** Form $f(s,\sigma_U(s),\sigma_C(s))$, solve stationary points and compute its full Jacobian, including policy gradients; report eigenvalues, basins of attraction where relevant, and robustness across grids/parameters. A fixed-control eigenvalue table is not a substitute.
10. **Numerical reproducibility.** Repeat deterministic toy cases, negative-beta invariance, naive-clipping counterexample, constant-control benchmark, and randomized admissible parameters before fitting the game to empirical state trajectories. Explicitly label whether a run is a *toy solution*, a calibrated counterfactual or a genuine empirical state estimate.

Global existence caveat. If an appropriately defined coupled policy/HJB map is a contraction on a closed admissible function space with compatible boundary conditions, that can suffice for uniqueness of a fixed point, but this is a **conditional assumption to verify**, not a theorem obtained merely from $\rho_i > 0$ and compact actions. Otherwise report *computed candidate MPE*, verification gaps and possible multiplicity. Mixed-strategy existence in a discretized finite stochastic game does not prove it survives mesh refinement to a smooth pure feedback solution.

8.7 Exact empirical-to-game mapping and calibration protocol

This was the largest unresolved conceptual bridge in prior drafts. The mapping is now **procedural and falsifiable**, though coefficients still require data:

(i) Semiconductor observation to game state. Freeze architecture a and essential layers $\mathcal{L}(a)$. For each quarter, construct $x_t^{\text{obs}} = \min_{\ell\in\mathcal{L}(a)} z_{\ell,a,t}$ with posterior/coder uncertainty and record the binding layer. For a differentiable alternative, $x_t^{(q)} = S_q(z_{1,t},...,z_{L,t})$ with the approximation error. DO NOT map $S_{\text{effective}}$ to slow x: it contains the immediate foreign-access shock. Compute x under both *fixed frontier* and *moving frontier* and show divergence. Weight different target architectures only when the objective explicitly defines multi-architecture exposure, with frozen demand weights.

(ii) Rare-earth observation to game state. For the same quarter, use the stage/grade-feasible LP under Chinese-material/processing exclusion to obtain $y_t^{\text{obs}} = \sum_g w_g d_{g,t}^{\text{CN0}}/D_{g,t} \in [0,1]$. Report y_t^{N1} as a distinct stress series; do not confuse raw K/D surplus with this bounded fraction. Align bloc membership, demand weights and network version before estimating annualized drift.

(iii) Control histories. Aggregate the frozen jurisdiction/layer legal series into u_t, using fixed target-layer dependence weights and observed scope; likewise aggregate element/product Chinese legal restrictions into c_t, using frozen pre-control Chinese-input dependencies. Because legal severity is ordinal, publish a binary event-model sensitivity and interval-valued restrictions; do not assume a half-step in legal category has half the economic effect of a ban. Version annualized quarterly control u_t, c_t and documentation for amendments/suspensions.

(iv) Pipelines and transition parameters. For a candidate κ_C, the discrete-time solution under constant quarterly u_q is $m_{q+1} = u_q + (m_q - u_q)e^{-\kappa_C \Delta}$; similarly n, $\Delta = 0.25$. Estimate κ from independent lagged investment/commissioning/capability response evidence where possible; if unidentifiable, register a grid of lag half-lives (e.g., 1, 2, 4, 8 years) and mark scenario. Conditional on piecewise-constant m_q, the *exact* one-quarter model transition for x is $x_{q+1} = x^*_{C,q} + (x_q - x^*_{C,q}) \exp[-(G_C(m_q) + \delta_C)\Delta]$, $x^*_{C,q} = G_C(m_q)/(G_C(m_q) + \delta_C)$, with a corresponding formula for y. For varying m within quarter integrate with a stable ODE solver; report approximation error of the piecewise-constant simplification. Fit $(g_C, \beta_C, \delta_C, \kappa_C)$ to observed x posteriors with measurement-error-aware likelihood or generalized method of moments; y analogously. If the minimum switches binding layers or a large plant creates an abrupt step, use a jump/observation-error extension or label the four-state approximation invalid for that interval.

(v) Parameter observability hierarchy. x, y are noisy **constructed states**; u, c are legal-intensity **constructed controls**; effort/capability effects θ_ℓ, η_ℓ, project hazards and $G(m)$ are **estimable only when observables suffice**; g, β, δ, κ may be **partially identified**; and $A_i, D_i, k_i, h_i, B_i, q_i, \rho_i$ are **mostly strategic/calibrated/scenario parameters**. Without independent outcome prices, costs and revealed preference variation, observed policy history generally cannot identify separate payoff coefficients. Explicitly label identifiability and posterior/prior sensitivity in the registry. When estimating payoffs by inverse game calibration, normalize a scale, account for policy noise and multiple equilibria, and show a profile objective ridge; no point estimate can be promoted to causal structural proof merely because an optimizer returned it.

(vi) Calibration objective and validation. Register an errors-in-variables criterion $\sum_q [(x_q - \hat{x}_q)^2/\sigma^2_{xq} + (y_q - \hat{y}_q)^2/\sigma^2_{yq}]$ + regularizer(θ), or a state-space likelihood including observation model, time aggregation and common shocks. Run multiple starting values and simulation-based calibration on synthetic data with known parameters to check recovery. Hold out post-event years where feasible; report observed and predicted drift, confidence intervals, sensitivity to frontier definitions and whether the simple dynamic law beats random-walk and autoregressive baselines. Reject any parameterization whose predicted state violates its empirical measurement range or sign constraints. For payoff scenarios, define transparent economically interpretable grids of coercion value, domestic cost, discount factor and strategic complementarity; never describe scenario-only restraints as an estimated U.S./China optimal foreign policy.

9. Counterfactuals and publication exhibits

The counterfactual analysis should distinguish **direct short-run access**, **delayed technology/plant adaptation** and **optimal strategic response**, rather than presenting one composite outcome as a policy verdict. Pre-register: (A) observed-policy baseline; (B) exogenous 25% higher/lower controls within observed scope; (C) unilateral control shock with opponent policy frozen for one year; (D) two-player endogenous response from the candidate MPE; (E) announced China-free industrial subsidy with contrac-

tually backed input/outtake; (F) lower/higher observable opacity holding geopolitical volatility constant; (G) adverse bottleneck/plant/country outage; (H) credible rapid licensing relaxation; (I) high versus low induced-adaptation elasticity, including a negative-effect scenario; (J) stricter and more flexible Chinese access/alternative architecture specifications. Do not compare policies on incompatible information sets or change the frontier between scenarios.

Outputs. Present quarterly probabilities of final magnet-service adequacy; stage bottleneck identity; fraction of supply dependent on Chinese feedstock; conditional rare-earth first passage; immediate semiconductor effective availability and slow domestic capability by layer; time paths of u, c, x, y, m, n; the U.S. and China's discounted *scenario payoffs* in normalized utility units; conditional alternative equilibrium paths; local stability/attraction diagnostics; and 10/50/90% parameter-sensitivity envelopes. The notion of a "leverage half-life" should be defined from a specific observed or modeled coercive benefit $L_U(t) = A_U u_t(1-x_t)$, conditional on a specified policy path. If the flow payoff is nonmonotone, no single half-life exists; report time to half the initial level only when well-defined. Treat projected bilateral game payoffs as normative modeling assumptions, not estimates of geopolitical welfare.

A publication-ready paper should ideally have these figures/tables: (1) dated policy chronology with revision/suspension statuses; (2) rare-earth value-chain schematic with stage concentration **source and year**; (3) registered project risk set and observed transition table; (4) fitted commissioning/ramp curves; (5) physical network material balances by element and product; (6) China-free service and N-1 worst-case heat maps; (7) first-passage cumulative probabilities with credible intervals; (8) vintage backtest skill relative to baselines; (9) legal-control severity by U.S./NL/JP layer; (10) domestic versus foreign versus effective Chinese semiconductor availability; (11) robust event-study coefficient plot; (12) opacity index and source-coverage panel; (13) opacity-hazard curve with turning-point uncertainty; (14) self-eroding benchmark versus myopic control; (15) feedback MPE policy maps and closed-loop trajectories; and (16) identifiability and sensitivity table. Do not manually modify plots after scripts generate them. Distinguish analytical/synthetic curves from empirical estimates with captions and legend text.

10. Source acquisition, chronology, provenance and ethics

10.1 Priority-source registry with concrete access constraints

The source list below is a **starting registry**, not a claim that any requested firm-level, technical-grade, licensing or company-contract dataset has already been downloaded. Dated events must be retrieved from authoritative archived artifacts whenever available. As-of forecasting needs **old versions**, so a contemporary live endpoint cannot substitute for historical publication history. Source IDs should be immutable; each version has HTTP URL/file path, publisher, title, content hash, original language, publication date, first seen date, retrieval time, access terms, column extraction map, and duplicate/adjudication log. The specific primary-policy and principal aggregate source URLs listed here were located in public official/publisher resources during this synthesis; granular extraction remains a future execution task.

Registry ID	Exact entry point	Target variables	Frequency, restrictions and fidelity
S01	IEA, *Rare Earth Elements* (2026): https://www.iea.org/reports/rare-earth-elements/executive-summary	Stage definitions, published baseline country shares, demand/project scenario benchmarks	Report year, scenario-based; not a project-by-project realized-output substitute
S02	IEA, *Global Critical Minerals Outlook 2026*: https://www.iea.org/reports/global-critical-minerals-outlook-2026/outlook	Announced-pipeline stage bottlenecks and mineral demand assumptions	Announced project scenarios, license/availability to check
S03	USGS MCS 2026: https://www.usgs.gov/publications/mineral-commodity-summaries-2026	Countries' mineral production, prices, trade and import reliance	Annual; does not directly measure every refining/magnet stage
S04	USGS data release: https://www.usgs.gov/data/mineral-commodity-summaries-2026-data-release	Machine-readable USGS CSV tables and metadata	Released Sept. 18, 2026; freeze excludes future corrections
S05	MOFCOM/GACC No.18, April 4 2025: https://english.mofcom.gov.cn/Policies/AnnouncementsOrders/art/2025/art_0dd87cbee7b045bf93fabe6ab2faceee.html%3A11	Legal products, classifications, export license scope and effective date	English for reference; archive authoritative Chinese instrument if ambiguity
S06	MOFCOM No.70, Nov. 7 2025: https://www.mofcom.gov.cn/zfxxgk/gkml/art/2025/art_b1bff63a6bdd412a99c84203dd30fd39.html	Suspension list and end date	Specific 55,56,57,58,61,62 suspension; no inferred rescission of No.18
S07	BIS December 2024 controls: https://www.bis.gov/press-release/commerce-strengthens-export-controls-restrict-chinas-capability-produce-advanced-semiconductors-military	SME/software/HBM/entity additions	Press release discoverability; authoritative regulatory text needed for ECCNs
S08	BIS January 2026 revision: https://www.bis.gov/press-release/department-commerce-revises-license-review-policy-semiconductors-exported-china	H200/MI325X case-by-case rules	Check Federal Register/EAR exact effective/legal language
S09	BIS Export Administration Regulations: https://www.bis.gov/regulations/ear/744	ECCN/end-use/end-user and section 744 interpretation	Version by Federal Register amendment and effective date; live site overwrites history
S10	Dutch government April 2025: https://www.government.nl/latest/news/2025/01/15/klever-export-controls-on-advanced-semiconductor-manufacturing-equipment-to-be-tightened	Controlled classes, national authorization, effective Apr 1	Dutch official legal instrument controls final scope; not blanket ban
S11	METI 2023 equipment controls: https://www.meti.go.jp/policy/anpo/law09-2.html	23 equipment items, amendments and effective July 23 2023	Japanese statute controls final detail; assess native text/translation

Registry ID	Exact entry point	Target variables	Frequency, restrictions and fidelity
S12	UN Comtrade API guide: https://uncomtrade.org/docs/un-comtrade-api/	Monthly/annual reporter-partner-product quantities and value	Free preview limited; full API authentication/tiers; HS revision crosswalk
S13	USITC DataWeb API: https://www.usitc.gov/applications/dataweb/api/dataweb_query_api.html	U.S. detailed import/export and partner/product data	Registered account/API key; vintage statistical revisions; access audit
S14	SEC EDGAR: https://www.sec.gov/edgar/search/	Public filings, ownership, capex, revenue and R&D	Filing date critical; company fiscal years differ; automated access rules
S15	WIPO PATENTSCOPE: https://www.wipo.int/en/web/patentscope	Patent-family dates, IPC/CPC technical activity	Patent ≠ verified qualified equipment capability; publication lag
S16	IEA critical-minerals data viewer: https://www.iea.org/data-and-statistics	Published scenario series	Some downloadable components subject to terms; units/scope must be checked
S17	Corporate stock-exchange and audited filings, national planning/permit agencies (case-specific)	Stage dates, FID, construction, commercial ramp, grants, technical grades	Manual registry with evidence URL and disclosure priority required
S18	WSTS / SIA market publications, https://www.wsts.org/	Aggregate semiconductor cycle/market size	Subscription and aggregation; cannot by itself identify advanced node/capability

Do not assert missing licensed private data can be downloaded. If WSTS, proprietary equipment technology, technical yield, project commercial contract terms, or customs-record detail is inaccessible, write a feasibility memo: public fallback, variables lost, resulting estimands, and whether the module must narrow scope. Scraped sources are not permission to redistribute personal or restricted data; replication packages should include code and lawful pointers/hash manifests when raw redistribution is prohibited. Do not publish internal company personal contacts or proprietary industrial process recipes.

10.2 Historical evidence protocol: two clocks plus source revisions

Every table has at least `valid_date` and `info_public_date`. Example: a facility starts commercial production on 2023-09-01; that fact is first publicly documented 2024-02-10. The full-information analytical panel may use September as the retrospectively observed event date, but a **December 2023 historical forecast must not contain that fact**. It must treat the project as unresolved as of December 2023. Any observed revision is retained as a new `record_version_id` with its own source hash/publication date; the earlier record stays immutable. If the earliest public date is unknown, exclude it from the relevant vintage forecast or conservatively place it after the earliest *verifiable* availability date and quantify the resulting loss of sample.

Link tables via explicit keys: source artifact → dated claim → facility/activity → project state → ownership/route/feedstock → scenario → LP output → forecast draw → figure. Every generated output should record the exact source and model registry. In the manuscript's main tables include *universe sizes and attrition*: announced projects, eligible projects, those with trustworthy stage capacity, those with technical composition, those with inferred routing, and those reaching operational confirmation. Missing industrial information is itself a substantive limitation, but missingness cannot be silently transformed into Chinese opacity.

11. Replication and computational implementation contract

11.1 Reproducible repository and clean build principle

The architecture below is what the eventual research project must build. This blueprint includes a separately packaged **minimal tested mathematical reference fixture** for limited components, but the full empirical pipeline is not yet implemented; the repository map is a contract, **not evidence that every listed module exists or passes**.

```
weaponizing-interdependence/
  README.md                 # scope, install, lawful data requirements
  pyproject.toml             # src-layout package + pinned, resolved dependencies
  uv.lock                    # generated and frozen on build machine
  Dockerfile                # verified base-image digest stored at release
  Makefile
  LICENSE  CITATION.cff  manuscript/
  configs/
    scope.yaml              # bloc, strategic grades, targets, thresholds
    data_sources.yaml       # official URLs + extraction obligations
    rare_earths.yaml        # process / ownership / loss scenarios
    semiconductors.yaml     # technology universe and legal crosswalk
    opacity.yaml            # 10-indicator rubric and validity rules
    game.yaml               # year units, parameters and scenario registry
    backtests.yaml          # origins, release/censor masks, scoring
    seeds.yaml              # independent reproducible RNG streams
  codebooks/
    project_stages.csv  process_coefficients.csv
    legal_policy_codes.csv  technical_targets.csv
    spoi_codebook.csv  grade_recipes.csv
  data/
    raw/                    # immutable, access-controlled source artifacts
    vintages/               # frozen as-of snapshots and release calendar
    interim/ processed/ fixtures/
  metadata/
    sources.csv hashes.csv adjudications.csv vintages.csv
    project_inclusion_log.csv parameter_registry.csv
  src/weaponized/
    io/{archive,vintages,provenance,schemas}.py
    rare_earths/{projects,transitions,ramps,materials,lp,resilience,simulate}.py
    semiconductors/{legal,crosswalk,access,evidence,capability,events}.py
    opacity/{obligations,events,coding,index,reliability}.py
    forecast/{origins,scoring,calibration,bootstrap}.py
```

```
    game/{primitives,benchmark,active_set,hjb,policy_iteration,stability}.py
    reporting/{figures,tables,manifest}.py
  scripts/{fetch,validate,build,fit,backtest,forecast,solve,report}.py
  tests/{unit,integration,property,regression}/
  outputs/{tables,figures,forecasts,equilibria,diagnostics}/
```

Exact interface contracts

```
load_asof(table_name: str, as_of: date) -> DataFrame
# Requires info_public_date <= as_of, with versioned de-duplication.

solve_material_flow(
    facilities, activities, input_output_coefficients,
    routes, demand, initial_inventories, policy_access,
    failed_groups, quarter, solver_config
) -> FlowResult
# Returns facility/commodity/quarter balance residuals, delivered services,
# bottlenecks, primal and dual diagnostics and integrity flags.

nminus1_resilience(network: Network, demand: Demand,
                   initial_exclusions: set[str],
                   candidate_suppliers: list[str]) -> DataFrame
# On every candidate outage, REBUILDS/re-solves full network LP.

simulate_projects(asof_snapshot, transition_model, ramp_model,
                  demand_scenarios, rng_streams, n_draws) -> DrawPanel
# Maintains common shocks, joint site risk, and individual project correlation.

spoi_daily(obligations, legal_events, source_events,
           codebook, start, end) -> DataFrame
# Ten explicit codes + NA reasons + daily validity; no future events.

estimate_semiconductor_states(evidence, rule_panel, target_version,
                              measurement_config) -> StatePosterior
# Returns separate z, f and a layers; can't conflate access with capability.

solve_stage_nash(mu, mc, k_u, k_c, q_u, q_c) -> list[ActionSolution]
# Enumerates all nine regimes and validates conditional best responses.

solve_feedback_game(state_grid, primitives, starts, tolerances) -> MPEResults
# Returns BOTH values, policies, residuals, deviation gaps, runtime,
# stability eigenvalues and grid-refinement audit; may return 'not solved'.

run_backtests(vintage_registry, endpoints, outcome_release_registry,
              scoring_config, baseline_set) -> BacktestReport
# Rejects future endpoints and outcomes not known by the assessment freeze.
```

Hard invariants: all state/service/capability/control values are within [0,1]; raw supply-to-demand ratios are allowed above 1; no activity output exceeds capacity or yield-adjusted feed; stage capacities cannot be added as equivalent commodities; only one supplier removed per N-1 scenario; source-vintage rule never violated; all policy controls have effective/publication/suspension dates; any mathematical

singularity or negative investment is an explicit error or constrained transform. Define floating-point tolerances in *dimensionful units* for balances (e.g. relative tolerance 10^{-8} times throughput plus an absolute 10^{-8} tonne-equivalent for toy fixtures), not one arbitrary bound for every unit scale.

11.2 Minimal locked configuration

```
freeze: '2026-10-11'
clock:
  empirical: quarter
  game: year
  quarter_year_fraction: 0.25
rare_earths:
  elements: [Nd, Pr, Dy, Tb]
  stages: [mine, separation, metal_alloy, magnet]
  service_scope: china_free
  n_minus_one: extra_supplier_group_after_china_removed
  targets: [0.50, 0.75, 0.90, 1.00]
  require_consecutive_quarters: 4
  freeze_grade_weights: true
  prohibit_unknown_imports_by_default: true
semiconductors:
  capability_state: literal_minimum_domestic_by_architecture
  access_function: residual_gap_approximation_with_recipe_robustness
  target_version: SET_BEFORE_CODING
  freeze_exposure_base_year: 2021
opacity:
  pillars: {disclosure: 0.40, predictability: 0.35, clarity: 0.25}
  event_memory_days: 90
  min_daily_indicators: 8
  min_valid_quarter_days_fraction: 0.80
  allow_headline_imputation: false
game:
  dynamics: exponential_growth_with_bounded_pipelines
  beta_sign_free: true
  state_grid: [17, 21]   # add 31 only if runtime feasible
  tolerance_hjb_scaled: 1.0e-8
  tolerance_best_response: 1.0e-7
backtests:
  rare_earth_origins: ['2016-12-31','2018-12-31','2020-12-31',
                       '2022-12-31','2024-12-31']
  rare_earth_horizons_years: [2, 4, 6]
  semiconductor_horizons_months: [6, 12, 24]
  block_future_endpoints: true
  require_outcome_publication_by_freeze: true
seeds:
  parameters: 101
  projects: 202
  process_yields: 303
  common_shocks: 404
  graphics: 505
```

`target_version:SET_BEFORE_CODING` is **intentionally unresolved**; publication-grade criteria require the user/researcher to register actual commercial technology requirements, and no responsible synthesis can invent them. Other unresolved empirical data such as route capacities, company relationships and material recipes must similarly be entered before full execution. Avoid placeholder version pins masquerading as a frozen release. On creation, choose Python 3.12 if dependencies/solver bindings support it, run a version resolver, commit a valid `uv.lock`, validate licenses, then freeze `PYTHON_IMAGE` by container registry digest and `UV_VERSION` as an exact verified version. `uvsync--frozen` must not modify the lock; the artifact must build without retrieving new sources during analysis. Existing Draft 5 versions were *proposals*, not a furnished checked dependency environment.

11.3 End-to-end DAG, provenance and execution

The data flow is `registersource→downloadormanualarchivewithoriginaltimestamp→hash→parse→validateunits,datesandkeys→adjudicateduplicatesandtechnicalclassifications→freezeas-ofsnapshot→estimateprojecthazardsandsemiconductoroutcomes→constructSPOI→simulateproject/flowforecasts→backtest→mapx,yandfitreducedstatetransitions→calibratescenario-onlypayoffs→solvecoupledHJBcandidates→computerobustness,exhibitsandmanuscript`. Scripts should be idempotent. In CI, data acquisition is mocked with immutable fixtures and there are **no live Web requests**; historical as-of sources are archived in separate signed, versioned collections. Every reproducibility run produces `run_manifest.json` with Git commit, Python/dependency and container digests, input and output source checksums, as-of cutoff, hypothesis registry version, data dictionary version, model settings, RNG seeds and generated figure/table hashes. Controlled raw data can have a non-redistributable manifest and access procedure rather than unauthorized redistribution.

Example production command contract (not presented as already executed):

```
uv sync --frozen
uv run pytest -q
uv run python scripts/validate.py --freeze 2026-10-11
uv run python scripts/build.py --freeze 2026-10-11
uv run python scripts/fit.py --config configs/rare_earths.yaml
uv run python scripts/fit.py --config configs/semiconductors.yaml
uv run python scripts/backtest.py --config configs/backtests.yaml
uv run python scripts/forecast.py --config configs/rare_earths.yaml
uv run python scripts/solve.py --config configs/game.yaml --multi-start
uv run python scripts/report.py --freeze 2026-10-11 --verify-manifest
```

Dockerized replication uses an exact pinned base image and correct src-layout installation (e.g. `uvsync--frozen--no-dev` with project installed, or explicitly `PYTHONPATH=/app/src` when executing standalone scripts). Every build is checked in a clean container and, before submission, on a second independent machine. An archive stored with the paper must contain legally shareable inputs or reproducible source retrieval instructions, code, tests, lockfile, config, manifest, and figure/table regeneration workflow. Merely naming `.py` modules in a PDF does not establish a runnable codebase.

11.4 Mandatory tests and failure semantics

Test	Exact test scenario	Required result	Current status
Bounded dynamics	Random $m, n, x, y \in [0,1]$, β negative and positive, admissible controls	Forward ODE stays [0,1]; boundary vector inward	Analytically proved; full solver unexecuted
Reduced-form steady state	Constant controls with analytic fixed points	ODE tends to $ge^{\beta u}/(ge^{\beta u} + \delta)$	Analytically derived; software future
Correct active-set Nash	$M_U = 1.2, M_C = 0.1, q_i = 0.8, k_i = 1$	$(1, 0.9)$, not independently clipped $(1, 1)$	Analytically verified; demonstrator fixture provided separately
One-sided coercion root	$A = v_0 = k = \rho = b = 1$	Interior $u \approx 0.4655712319$, below myopic 1	Numerically verified for this document
Element/grade material conservation	Positive yield and shared activity	Output never exceeds resource input and nameplate	Conceptually specified; full LP model pending
Supplier outage	Toy A/B oxide/alloy/magnet fixture	Base served 1, remove A 0.5, remove B 4/7, N1=0.5	Analytically verified; fixture provided separately
Ownership cascade	A supplies B downstream	Removing A changes B's achievable production	Full network test required, not yet performed
Uncapped/raw versus capped service	75 / 70 potential	Raw 1.0714286, served fraction 1.0	Analytically verified
Opacity arithmetic	All 10 indicators, two synthetic vectors	57.4167 and 16.75	Recomputed; synthetic only
Opacity nonevent	90 days no new adverse event	O4/O6/O10 = 0 under prescribed windows	Test rule specified; not yet run against events
Historical vintage	Record valid 2023, first public 2024	2023-origin forecast rejects record	Test contract specified; full data absent
Future endpoint	Origin 2020 +6 years	2026-12-31 endpoint not scoreable Oct 11 2026	Schedule fixed; software validation pending
MPE verification	Two sparse HJB solves / multi-start / deviation gaps	Residual and KKT checks pass and mesh refinement stable	**Not executed**
Forecast calibration	Vintage-eligible project events	Proper scores beat or honestly fail baselines	**Not executed**
Reproducible release	Build exact source snapshot in clean pinned container	Every manuscript exhibit reproduced	**Not executed**

The acceptance convention is intentionally asymmetric: a proved symbolic identity is not a validated empirical model; an executed toy script is not a full research reproduction; and a mathematically correct active-set stage game does not prove a continuous-state feedback equilibrium. Any failure produces a logged `FAIL` or `NOTYETTESTED`, not a silently green completion report.

12. Staged implementation roadmap and decision gates

Gate 1 — scientific feasibility and scope (first). Freeze technology target envelopes; define bloc B/ownership country scenarios; choose magnet grades/demand and product units; verify primary legal timeline and baseline IEA/USGS statistics; establish a preliminary inclusive universe of candidate projects and identify missing public documents. Deliver a one-page evidence-availability dashboard by

stage/layer. A project that cannot distinguish actual operation from announcements cannot proceed to probabilistic capacity inference without major caveats.

Gate 2 — physical project registry and network (weeks 2–10, schedule contingent). Manually adjudicate project/stage dates and source-vintage versions; build product/element conversion tests and physically feasible routes; construct and validate base/China-removal/N1 network examples; add conservative ambiguity bounds. Produce an audit showing how gross announced capacity differs from *feasible* final magnet service and which missing stage blocks it. Do not fit hazards before validating risk-set definition and censoring.

Gate 3 — semiconductor legal and functional evidence (in parallel). Freeze control-item and technology-layer maps, collect firm/product exposure and imports, and independently code actual indigenous milestones. Construct balanced and unbalanced panels with missingness reasons. Pre-trend and treatment-overlap checks are gating conditions for strong causal conclusions; if unsuitable, frame estimates as associative descriptive panel evidence or focus on shorter event-window descriptive access changes.

Gate 4 — opacity codebook reliability. Establish release obligations and policy event universe; perform coder training; reach reliability target or revise rules *without conditioning on project outcomes*. Only then produce headline quarterly SPOI. If too few obligations exist for certain indicators, disclose limited coverage and narrow scope instead of fabricating apparent regulatory transparency.

Gate 5 — estimates and forecasts. Estimate multistate transition probabilities with shrinkage; generate correlated commissioning, ramp, yield, stock and demand draws; solve network for all registered outages; run origin-valid backtests with proper scores and compare alternatives. Show model inadequacy if the forecast systematically understates failures or exceeds plausible IEA pipeline scale without technical evidence.

Gate 6 — measured dynamics and theory. Map layer and network observations into x, y with uncertainty; fit or bound g, β, δ, κ. If reduced states are too discontinuous or poorly identified, expand state dimension or present the differential game as *partial-identification scenarios*. Derive analytical properties and run the full coupled HJB from multiple starts; report `NOVERIFIEDPUREMPEFOUND` honestly if the solver cannot establish a reliable result.

Gate 7 — paper writing, independent review and submission. Freeze code and data; review all policy/legal and technical claims from original sources; write Introduction, Institutional Background, Prior Literature, Research Design, Mineral Results, Semiconductor Results, Opacity Results, Game & Counterfactual Results, Robustness, Limitations, Conclusion and complete references. A separate technical appendix contains proofs, schemas, figures, source eligibility rules, tests and reproducibility manifest. Invite an economist familiar with dynamic games, an operations-research/material-processing specialist, and a semiconductor controls specialist to challenge separate components before submitting.

Publication contingency if data access is poor. Rather than fabricate a symmetric full structural result, retain a high-quality **modular paper**: project-based rare-earth backtests and physically grounded forecasting as principal empirical result; documented control chronology and carefully qualified semiconductor adaptation evidence as independent second study; opacity measurement as exploratory; and a scenario/calibrated reciprocal model whose identified and unidentifiable parameters are visibly distinct. The paper remains intellectually coherent with this narrower evidentiary claim; failure to collect confidential reactor yields or firm revenues is not fixed by increasing mathematical complexity.

13. Annotated literature and source bibliography

The bibliography below uses author–date conventions with DOI or an official link, and tracks **source status**. For the core papers and primary-policy entries explicitly checked against publisher/agency pages during this synthesis, mark **P** (publisher/agency entry checked). **D** indicates a plausible citation retained from the uploaded mature drafts whose bibliographic metadata or exact claim correspondence still needs manuscript-stage verification; it must *not* be promoted to "fully verified APA7" until the final audit is carried out. **M** denotes a data/manual entry where the relevant table or live information still needs extraction by the research team. DOI metadata may change as accepted papers receive final issue/pages. Scholarly source type is identified: peer-reviewed article, working paper, book, or official document. This is a usable working bibliography, not a fabrication of a completed source-to-claim replication audit.

13.1 Foundational network power, resource competition and coercion

[**R01 | P, journal**] Farrell, H., & Newman, A. L. (2019). Weaponized interdependence: How global economic networks shape state coercion. *International Security, 44*(1), 42–79. https://doi.org/10.1162/isec_a_00351. **Contribution:** Foundational network-topology/chokepoint concept. **Difference:** Does not deliver this paper's joint industrial project-level forecasts and adaptation-conditioned feedback equilibrium. **Use:** theoretical introduction and precise novelty boundary.

[**R02 | P, working paper**] Clayton, C., Maggiori, M., & Schreger, J. (2024; revised April 2025). *A theory of economic coercion and fragmentation* (NBER Working Paper No. 33309). National Bureau of Economic Research. https://doi.org/10.3386/w33309. **Contribution:** Formal economic coercion and integration–security trade-off, with sector concentration and substitute availability. **Difference:** The intended empirical innovation here is time-resolved technology and material adaptation with physical completion uncertainty, not simply reconfirming fragmentation costs. **Use:** core theoretical competitor.

[**R03 | P, journal**] Woods, D. (2026). The "black box" strategy: China's opacity signaling game in rare earths (REES). *Asian Review of Political Economy, 5*, Article 5. https://doi.org/10.1007/s44216-026-00073-z. **Contribution:** Pooling/nondisclosure signaling and stochastic diversification simulation; simulation assumptions are explicitly illustrative. **Difference:** Present project constructs an auditable observable-policy index, estimates association with **actual** project transitions and does not treat hypothetical Monte Carlo probabilities as forecasts validated against completed plants. **Use:** closest opacity precedent; distinguish observable uncertainty from presumed intent.

[**R04 | P, journal**] Zou, B., Poncin, S., & Bertinelli, L. (2022). The U.S.–China supply competition for rare earth elements: A dynamic game view. *Environmental Modeling & Assessment, 27*, 883–900. https://doi.org/10.1007/s10666-022-09819-4. **Contribution:** Published rare-earth resource-competition differential game, including open-loop and Markovian strategic behavior. **Difference:** The new game links *different* reciprocal chokepoints and estimated adaptation; it must not claim to be the first rare-earth differential game. **Use:** closest resource-game comparison, especially equilibrium concepts.

[**R36 | P, accepted journal manuscript**] Liu, J., Rotemberg, M., & Traiberman, S. (2026). Sabotage as industrial policy. *The Economic Journal*, Advance online publication, Article ueag111. https://doi.org/10.1093/ej/ueag111. **Contribution:** Formal conditions under which foreign productivity impairment can have non-monotone terms-of-trade and welfare effects. **Difference:** Reciprocal future adaptation and the *destruction of coercive leverage* over time, rather than only the contemporaneous

industrial-policy welfare consequence. **Use:** theory positioning and caution on treating restrictions as automatically good or bad.

13.2 Semiconductor adaptation: disagreement is substantive

[**R05 | P, journal**] Anwar, S., Hu, B., Luan, Q., & Wang, K. (2024). Export controls and innovation performance: Unravelling the complex relationship between blacklisted Chinese firms and U.S. suppliers. *The World Economy, 47*(7), 2995–3033. https://doi.org/10.1111/twec.13570. **Contribution:** Matching/difference-in-differences evidence of innovation responses among blacklisted Chinese firms and impacts on U.S. suppliers. **Limit:** Innovation indicators and treatment selection do not automatically demonstrate qualified substitution at leading technology layers. **Data caveat:** underlying CSMAR data subject to license per publisher.

[**R06 | D, journal**] Hu, H., Yang, S., Zeng, L., & Zhang, X. (2024). U.S.–China trade conflicts and R&D investment: Evidence from the BIS entity lists. *Humanities and Social Sciences Communications, 11*, Article 829. https://doi.org/10.1057/s41599-024-03369-8. **Contribution:** Firm R&D/input response to restrictions. **Limit:** R&D is not functioning semiconductor capability. **Action:** confirm exact authors/metadata and its effect definitions with publisher PDF before final manuscript.

[**R07 | P, journal**] Hayakawa, K., & Yang, C.-H. (2026). Technology sanctions and firm R&D: Evidence from the U.S.–China trade dispute. *Economics & Politics, 38*(2), 406–419. https://doi.org/10.1111/ecpo.70033. **Contribution:** A matching analysis that does **not** find a significant positive R&D increase for treated firms. **Interpretation:** sign of induced effort must be estimated rather than structurally assumed positive. **Limit:** Firm-level sample and R&D do not directly measure foundry yield or equipment qualification.

13.3 Investment uncertainty, project duration and causal designs

[**R08 | D, journal**] Bloom, N. (2009). The impact of uncertainty shocks. *Econometrica, 77*(3), 623–685. https://doi.org/10.3982/ECTA6248. **Use:** macro uncertainty/shock controls; not itself a measure of strategic nondisclosure.

[**R09 | D, journal**] Gulen, H., & Ion, M. (2016). Policy uncertainty and corporate investment. *The Review of Financial Studies, 29*(3), 523–564. https://doi.org/10.1093/rfs/hhv050. **Use:** irreversibility/uncertainty and investment response; retain confounding risks when applying to REEs.

[**R10 | D, journal**] Handley, K., & Limão, N. (2015). Trade and investment under policy uncertainty: Theory and firm evidence. *American Economic Journal: Economic Policy, 7*(4), 189–222. https://doi.org/10.1257/pol.20140068. **Use:** trade-policy uncertainty and firm investment; check the transfer from tariff uncertainty to licensing ambiguity.

[**R11 | D, book**] Dixit, A. K., & Pindyck, R. S. (1994). *Investment under uncertainty.* Princeton University Press. **Use:** real-option delay intuition. **Limit:** no direct rare-earth project-hazard parameter estimated in this book.

[**R12 | D, journal**] Cox, D. R. (1972). Regression models and life-tables. *Journal of the Royal Statistical Society: Series B (Methodological), 34*(2), 187–202. https://doi.org/10.1111/j.2517-6161.1972.tb00899.x. **Use:** cause-specific duration-model robustness, proportional-hazards diagnostics.

[**R13 | D, journal**] Fine, J. P., & Gray, R. J. (1999). A proportional hazards model for the subdistribution of a competing risk. *Journal of the American Statistical Association, 94*(446), 496–509. https://doi.org/10.1080/01621459.1999.10474144. **Use:** cumulative incidence robustness when cancellation competes with commissioning; not a substitute for the full transition simulator.

[**R14 | D, journal**] Callaway, B., & Sant'Anna, P. H. C. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics, 225*(2), 200–230. https://doi.org/10.1016/j.jeconom.2020.12.001. **Use:** heterogeneous cohort-specific treatment effects.

[**R15 | D, journal**] Sun, L., & Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics, 225*(2), 175–199. https://doi.org/10.1016/j.jeconom.2020.09.006. **Use:** avoiding misleading TWFE event-time averages.

[**R16 | D, working paper**] Callaway, B., Goodman-Bacon, A., & Sant'Anna, P. H. C. (2024). *Difference-in-differences with a continuous treatment* (NBER Working Paper No. 32117). https://doi.org/10.3386/w32117. **Use:** thinking about dose/exposure when a well-meaning continuous treatment measure is defensible. Ordinal rule category scores by themselves are not naturally continuous dosages.

[**R17 | D, journal**] Gneiting, T., & Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association, 102*(477), 359–378. https://doi.org/10.1198/016214506000001437. **Use:** CRPS/log-score/probability forecast calibration; explicit error score definition.

[**R18 | D, journal**] Brier, G. W. (1950). Verification of forecasts expressed in terms of probability. *Monthly Weather Review, 78*(1), 1–3. **Use:** binary forecast scoring; time-to-event scoring must address censoring.

13.4 Game theory, computation and nonlinear tipping

[**R19 | D, journal**] Maskin, E., & Tirole, J. (2001). Markov perfect equilibrium: I. Observable actions. *Journal of Economic Theory, 100*(2), 191–219. https://doi.org/10.1006/jeth.2000.2785. **Use:** equilibrium concepts; not a global existence theorem for the specialized continuous-state game.

[**R20 | D, journal**] Fink, A. M. (1964). Equilibrium in a stochastic n-person game. *Journal of Science of the Hiroshima University, Series A-I, 28*, 89–93. **Use:** stationary mixed equilibria in finite discounted stochastic games; the pure continuous feedback game remains conditional.

[**R21 | D, journal**] Pakes, A., & McGuire, P. (2001). Stochastic algorithms, symmetric Markov perfect equilibrium, and the "curse" of dimensionality. *Econometrica, 69*(5), 1261–1281. https://doi.org/10.1111/1468-0262.00241. **Use:** computational caution and MPE methods.

[**R22 | D, book**] Bardi, M., & Capuzzo-Dolcetta, I. (1997). *Optimal control and viscosity solutions of Hamilton–Jacobi–Bellman equations*. Birkhäuser. https://doi.org/10.1007/978-0-8176-4755-1. **Use:** PDE/state constraint/upwinding mathematics. Verify edition metadata before final BibTeX.

[**R23 | D, journal**] Granovetter, M. (1978). Threshold models of collective behavior. *American Journal of Sociology, 83*(6), 1420–1443. https://doi.org/10.1086/226707. **Use:** individual adoption thresholds generating aggregate regimes; analogy, not data from mineral projects.

[**R24 | D, journal**] Watts, D. J. (2002). A simple model of global cascades on random networks. *Proceedings of the National Academy of Sciences, 99*(9), 5766–5771. https://doi.org/10.1073/pnas.082090499. **Use:** distinguishes local cascade thresholds from ordinary physical loss propagation.

[**R37 | D, journal**] Schelling, T. C. (1971). Dynamic models of segregation. *Journal of Mathematical Sociology, 1*(2), 143–186. https://doi.org/10.1080/0022250X.1971.9989794. **Use:** path dependence and nonlinear collective composition; methodological analogy only.

[**R38 | D, journal**] Arthur, W. B. (1989). Competing technologies, increasing returns, and lock-in by historical events. *The Economic Journal, 99*(394), 116–131. https://doi.org/10.2307/2234208. **Use:** positive feedback and technological lock-in, motivating but not proving industrial REE tipping.

13.5 Official baseline, legal instruments and data entries

[**R25 | P, agency report**] International Energy Agency. (2026). *Rare earth elements*. https://www.iea.org/reports/rare-earth-elements/executive-summary. **Use:** primary chain taxonomy, 2024 production-concentration baselines, 2035 announced capacity comparators. **Constraint:** its pipeline is a forecast/scenario, not realized operating output.

[**R26 | P, federal news / linked legal instrument**] U.S. Bureau of Industry and Security. (2024, December 2). *Commerce strengthens export controls to restrict China's capability to produce advanced semiconductors for military applications*. https://www.bis.gov/press-release/commerce-strengthens-export-controls-restrict-chinas-capability-produce-advanced-semiconductors-military. **Use:** 24 equipment classes, three software controls, HBM; use linked Federal Register for binding ECCNs.

[**R27 | P, agency release**] U.S. Geological Survey, National Minerals Information Center. (2026). *Mineral commodity summaries 2026*. https://doi.org/10.3133/mcs2026. **Data release:** https://www.usgs.gov/data/mineral-commodity-summaries-2026-data-release. **Use:** official mineral statistics and reproducible data baseline. **Constraint:** incomplete all-stage technical manufacturing coverage.

[**R28 | P, Dutch government**] Government of the Netherlands. (2025, January 15). *Klever: Export controls on advanced semiconductor manufacturing equipment to be tightened*. https://www.government.nl/latest/news/2025/01/15/klever-export-controls-on-advanced-semiconductor-manufacturing-equipment-to-be-tightened. **Use:** April 2025 expansion of national **authorization**, not blanket ban.

[**R29 | P, Japanese government**] Ministry of Economy, Trade and Industry. (2023). *Export control amendments on semiconductor manufacturing equipment* [Official amendments and implementation notices]. https://www.meti.go.jp/policy/anpo/law09-2.html. **Use:** 23 classes and legal effective date. **Action:** final manuscript verify Japanese legal title and translation rather than treating this English description as the statute.

[**R30 | P, federal agency**] U.S. Bureau of Industry and Security. (2026, January 13). *Department of Commerce revises license review policy for semiconductors exported to China*. https://www.bis.gov/press-release/department-commerce-revises-license-review-policy-semiconductors-exported-china. **Use:** case-by-case standards for specified H200/MI325X-class export license requests.

[**R31 | P, Chinese government**] Ministry of Commerce of the People's Republic of China & General Administration of Customs. (2025, April 4). *Announcement No. 18 of 2025: Export control on some medium and heavy rare-earth-related items*. https://english.mofcom.gov.cn/Policies/AnnouncementsOrders/art/2025/art_0dd87cbee7b045bf93fabe6ab2faceee.html%3A11. **Use:** item-specific scope/licensing; official Chinese version authoritative if translation ambiguity.

[**R32 | P, Chinese government**] Ministry of Commerce of the People's Republic of China & General Administration of Customs. (2025, November 7). *Announcement No. 70 of 2025: Suspension of specified announcements*. https://www.mofcom.gov.cn/zfxxgk/gkml/art/2025/art_b1bff63a6bdd412a99c84203dd30fd39.html. **Use:** six suspended instruments (55,56,57,58,61,62), current-through 10 November 2026. **Constraint:** does not automatically determine status of every separate older export rule.

[**R33 | P, official data guide**] U.S. International Trade Commission. (n.d.). *DataWeb API user guide*. https://www.usitc.gov/applications/dataweb/api/dataweb_query_api.html. **Use:** official U.S. trade data extraction; requires DataWeb account and valid API key.

[**R34 | P, official data guide**] UN Comtrade. (n.d.). *UN Comtrade API*. https://uncomtrade.org/docs/un-comtrade-api/. **Use:** bilateral product trades with publication/version awareness; free API preview restricted and fuller extraction may require subscription.

[**R35 | P, official database**] World Intellectual Property Organization. (n.d.). *PATENTSCOPE*. https://www.wipo.int/en/web/patentscope. **Use:** published patent families and dates; not directly evidence of industrial qualified output.

[**R39 | M, institutional data**] U.S. Securities and Exchange Commission. (n.d.). *EDGAR filings*. https://www.sec.gov/edgar/search/. **Use:** ownership/financial statements, reported capacities, source and filing timestamps.

[**R40 | P, official outlook**] International Energy Agency. (2026). *Global Critical Minerals Outlook 2026: Outlook*. https://www.iea.org/reports/global-critical-minerals-outlook-2026/outlook. **Use:** annual published global stage pipeline and demand scenario benchmark.

Closest-source differentiation table

Existing contribution	What this blueprint attempts that source does not settle	Needed evidence before claiming success
Farrell & Newman (2019)	Computes endogenous destruction of choke leverage through real adaptation	Verified model and physical/access state paths
Zou, Poncin & Bertinelli (2022)	Joins mineral and semiconductor vulnerabilities with stage/layer empirical evidence	Two independent empirical modules and game bridge
Woods (2026)	Replaces illustrative opacity parameters with an independently coded observable index and investment transition estimates	Valid SPOI coding and adequate project-event sample
Anwar et al. (2024) / Hayakawa & Yang (2026)	Adjudicates mixed findings by separating effort, usable foreign access and verified domestic capability	Technical milestone data plus legal/firm causal design
IEA (2026)	Produces transparent probability distributions for commissioned, routed, usable supply and vintage backtests	Actual project/element/route histories; validated demand
Clayton et al. (2024/25)	Quantifies policy adaptation and future leverage in a more granular reciprocal mechanism	Honest scenario strategic payoffs and HJB verification

Bibliography status: high-priority journal titles/publisher claims and official policy entries were cross-checked selectively; **the bibliography is not yet fully independently audited source by source**.

A publication submission must resolve every D/M flag with publisher metadata, source-to-claim reading, exact legal-effective-date records and an exported `references.bib`. Unknown or dubious citations from the eleven-page summary have deliberately **not** been carried forward merely to pad the reference count.

14. Final audit: exact completion boundary and release blockers

A request to "make this complete" must distinguish **a coherent master design** from **an implemented, empirically publishable study**. A master document may fully specify methods while honestly lacking observed data and solved model coefficients; an empirical results section cannot be completed from unsupported guesses. Some pre-data methodological choices are intentionally *registered but require domain-specific evidence*: target performance envelopes, verified magnet recipes, ownership/capacity and supplier-route data, accessible license statistics, payoffs. The audit therefore lists **specified/proved**, **partially specified**, and **unverified/execution required** rather than certifying everything with an unchecked "complete" stamp.

Requirement from drafts/requests	Where resolved here	Blueprint assessment	Execution verification
Both original chokepoints remain separate, equally substantive	§§1, 3–7, 8	**Specified**	Datasets not built
No independent AI ecosystem game	§§0–1	**Specified**	N/A
As-of October 11, 2026 policy chronology; Chinese suspended vs active rules	§§2,10,13	**Specified; official sources located**	Legal-code item-by-item audit outstanding
Rare-earth feedstock → separation → metal/alloy → magnet process balances	§3.3	**Specified as LP**	Industrial recipes/routes/data missing; full LP not yet run
China-free versus N-1, one supplier removed across stages	§§3.4–3.5	**Specified, synthetic example verified**	Real network/outage fixture still required
Raw coverage may exceed 1; capped final service/game y bounded	§§3.4, 5, 8	**Reconciled**	Real supply/demand extraction pending
Commissioning, competing risks, ramp, common shocks, forecasts	§4	**Specified**	Estimates/forecast backtests unrun
Avoid invalid future historical backtest endpoints	§§4.3, 6.6	**Corrected**	Per-variable publication masks unbuilt
Distinguish threshold/viability from genuine bifurcation	§5	**Derived, conditional**	Feedback parameter unestimated
Semiconductor legal exposure vs foreign access vs actual domestic capability	§6	**Specified**	Technical target and evidence panel not frozen
Negative policy effect with nonnegative effort and bounded capability	§§6.4, 8.2	**Mathematically repaired**	Empirical θ, β unestimated
Strategic opacity 10 indicators, event/no-event rules, missingness, two examples	§7	**Operationally specified**	Historic coding and double-coder reliability unrun
Correct Markov four-state game and inward state dynamics	§8.2	**Derived and proved**	Numerical solver unbuilt
Exact HJB gradient/control FOCs and all nine constrained Nash regimes	§§8.4–8.6	**Derived, stagewise**	Grid verification unrun
Do not mistake pair-clipped interior solution for Nash	§8.4	**Corrected with counterexample**	Fixture verification included

Requirement from drafts/requests	Where resolved here	Blueprint assessment	Execution verification
Distinguish fixed-policy stability, action adjustment and full feedback stability	§8.6	**Distinguished**	Full closed-loop eigenvalues unknown
Honest dynamic-game equilibrium existence, uniqueness and multiplicity	§§8.4–8.6	**Conditional**	No global pure MPE certified
Constant-control self-eroding benchmark	§8.5	**Derived and numerically checked**	No unrestricted optimum claimed
Exact physically/evidentially grounded game state maps	§8.7	**Operational formulas specified**	Fit and adequacy testing outstanding
All strategic payoff parameters genuinely identified from data	§8.7	**Not feasible to certify ex ante**	Mostly scenario-only, partial identification must be assessed
Full source hierarchy, release clocks, variable dictionary and provenance	§§2–4, 6, 10	**Specified**	Historical snapshots/rights pending
Runnable *complete* empirical repository, pinned environment, all CI tests	§11	**Design contract only**	**Not delivered / not executed**
Full mathematically validated software HJB & project LP	§§8,11	**Algorithms specified; partial toy fixture**	**Not implemented**
Verified complete APA7 bibliography and source-to-claim audit	§13	**Working 35+ source registry**	D/M entries need final checks
Publication-quality estimated coefficients, probabilities, equilibrium comparisons	§§4,6–9	**Not yet research output**	**Not performed**

Requirements that remain genuinely outstanding

1. **Full implementation and validation.** The end-to-end material-flow LP, adaptive capability inference, cause-specific competing-risk estimator, as-of source loader, daily opacity event processor, four-dimensional coupled HJB solver, and reproducible manuscript builder have not been coded/executed together. The optional demonstrator tests only specified *toy* mathematical components and cannot certify these modules. No real network or value function has been solved.
2. **Source and technical scope verification.** Company-by-company historical stage inputs, elemental compositions, process yields, facility ownership, foreign feedstock, exact magnet-grade recipes, commercial throughput and accessible license data must be independently assembled. Critical semiconductor architecture/frontier targets must be frozen with expert input before a $z = 1$ score is meaningful. Source/license permissions constrain replication availability.
3. **Empirical estimability and true forecast backtests.** The number of rare-earth projects reaching each state, amount of independent 2016–2024 public-vintage material, length of post-2022 semiconductor exposure, and quality of non-treated comparisons cannot be presumed sufficient. If underpowered, simplify rather than report spurious precision. Backtesting cannot be certified without outcomes and timestamps.
4. **Game identification and verified equilibria.** Future state adaptation functions can be estimated or bounded, but national strategic payoff magnitudes are unlikely to be point-identified from a handful of restrictions. Pure feedback MPE existence and convergence are not proved for the calibrated full game merely by showing an action-level Nash fixed point. Report parameter-region and numerical equilibria *conditionally* after implementation.

5. **Final citation and external review.** Sources flagged D/M in Section 13 need original-document/bibliographic verification. Unpublished company announcements and claims about technologies need at least corroborating admissible sources. Publication-ready copy editing and specialist review remain.

Explicit certification decision

MASTER SYNTHESIS / MATHEMATICAL DESIGN: coherent and substantially more complete than any single provided draft, with the principal historical mathematical contradictions fixed and operational questions made explicit.

COMPLETE, EXECUTED IMPLEMENTATION-READY REPLICATION PACKAGE: NOT YET VERIFIED AND NOT CLAIMED. The full code, industrial dataset, model estimates and manuscript outputs do not yet exist in this deliverable.

PUBLISHABLE EMPIRICAL RESEARCH PAPER: NOT COMPLETE. Publication status depends on actual evidence, successful historical validation, credible identification, tested numerical equilibria and references. No manuscript can honestly announce the future probability of Chinese supply leverage ending by a given year without those calculations.

What success will look like: the corresponding named modules in the acceptance matrix will change from `NOTEXECUTED` to `PASS`, with source digests, public data manifests, figure hashes, independent numerical tolerances and cross-machine reproduction logs. Until then, the conclusion is a **carefully qualified master research blueprint**, not a certified empirical result.